

Inverse Problem Approach for Non-Perturbative QCD: Applications

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Abstract

We establish the first rigorous inverse problem framework for determining non-perturbative quantities of quantum chromodynamics, such as hadron decay constant and spectroscopy, thereby enabling mathematically precise quantification without reliance on phenomenological assumptions. By reformulating correlation functions through dispersion relations with operator product expansion inputs, we directly compute spectral densities – circumventing quark-hadron duality and Borel transformations. The precision is systematically improvable via higher-precision inputs. Critically, our approach provides fundamental new insights into the interplay between perturbative and non-perturbative QCD regimes, exemplified by revealing emergent resonance structures whose existence is validated through τ -decay measurements and exact Weinberg sum rules. We obtain the decay constants of a_1 and predict the branching ratios: $\mathcal{B}(\tau \rightarrow a_1(1260)\nu_\tau) = 0.184 \pm 0.032$ and $\mathcal{B}(\tau \rightarrow a_1(1640)\nu_\tau) = 0.015 \pm 0.004$, which can be directly confronted with future data. These results present a new paradigm for hadronic spectroscopy that will be highly beneficial for deepening our understanding towards the diverse facets of the strong interaction.

(Unique solutions for (1) pure perturbative input and (2) perturbative and condensate input.)

(How do resonances reproduced via OPE?)

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Contents

1	Introduction	2
2	Theoretical framework	4
2.1	Correlation function and Dispersion Relation	4
2.2	Inverse Problem	5
3	Numerical Results	8
3.1	Results under Tikhonov Regularization	8
3.2	Uncertainty Analysis	12
3.3	Convergence of Solutions and Precision Improvement	15
4	Physical Discussions	15
4.1	$\Pi_g(q^2)$ v.s. $\Pi_q(q^2)$	15
4.2	Quark-Hadron Duality	17
4.3	Vacuum Condensates	20
5	Summary and Outlook	23
A	Proof of Uniqueness and Instability of Inverse Problem with a Pole	23

1 Introduction

Quantum chromodynamics (QCD) stands as the fundamental gauge theory governing strong interactions, with its cornerstone asymptotic freedom [1, 2] enabling rigorous perturbative treatment of short-distance dynamics. However, this theoretical framework confronts profound challenges in the long-distance region, where non-perturbative effects dominate. Key among these are deciphering confinement mechanisms, hadronic mass spectra and resonance structures, as well as understanding the decay characteristics of various hadronic states and their scattering amplitudes. These non-perturbative features are indispensable for addressing many fundamental problems in particle physics, such as high-precision search for new physics, resolving the muon g-2 anomaly, and precision determinations of W-boson mass [3–7]. Therefore, the essential physical problem is: how to identify a correct low-energy description of QCD, which is the main objective in this work.

There are many non-perturbative methods to describe the low-energy physics, all of which have their own strength and weakness. Lattice QCD, as a first-principles approach, provides unparalleled precision for ground-state hadronic properties [8, 9]. Despite this capability, it faces inherent challenges in resolving higher excited states, resonant spectra, and broad decay widths [9, 10]. The results of Dyson-Schwinger equation depend on the approximation for solving the corresponding Bethe-Salpeter equations [11], such as the rainbow-ladder truncation. Fortunately, the well-defined correlation functions in quantum field theory obey dispersion relations based on physical causality. These relations are fundamental identities that bridge perturbative ultraviolet (UV) physics and non-perturbative infrared (IR) dynamics:

$$\boxed{\text{Calculable}} \Pi(q^2) - \Pi(\sigma_0) = \frac{(q^2 - \sigma_0)}{\pi} \int_0^\infty \frac{\boxed{\text{To be solved}} \text{Im}\Pi(s) ds}{(s - \sigma_0)(s - q^2)}. \quad (1)$$

The correlation functions on the left-hand side (L.H.S.) of Eq. (1) can be well treated through the perturbative QCD (pQCD) and operator production expansion (OPE) [12, 13]. The $\text{Im}\Pi(s)$ on the right-hand side (R.H.S.) of Eq. (1) contains low-energy physics which can not be calculated perturbatively. Therefore, we are led to ask how the dispersion relation can provide a pathway to identify the correct low-energy description from reliable pQCD and OPE calculations.

While dispersion relation find practical application in QCD sum rules (QCDSR) [12, 13], their implementation relies on the quark-hadron duality hypothesis – a theoretical construct introducing uncontrolled systematic uncertainties. This approach maps the non-perturbative spectral density $\text{Im}\Pi(s)$ onto an adjustable phenomenological threshold parameter s_0^H , with the physics of low-energy QCD unsolved. More critically, the duality assumption itself may introduce unquantifiable errors [14, 15]. On the other hand, many artificial and discretionary optimization and stability criteria are adopted to determine the corresponding parameters, which risks substantial systematic errors [16, 17]. In particular, the QCDSR-derived decay constants f_H have monotonic dependence on the continuum threshold s_0^H , where $\partial f_H / \partial s_0^H > 0$ [18, 19], which could introduce bias towards "good" results owing to the artificial adjustments of parameters.

The inverse problem approach provides a paradigm-shifting framework that rigorously reconstructs non-perturbative dynamics through fundamental dispersion relations [15, 20, 21], as depicted in Fig. ???. Its core methodology formulates the extraction of low-energy spectral functions ($|s| < \Lambda$) from high-energy inputs ($|s| > \Lambda$ via pQCD + OPE) as a mathematically well-defined inverse problem without model dependency, where Λ denotes a sufficiently large scale demarcating perturbative and non-perturbative regimes. However, this inverse problem exhibits inherent instability and thus qualifies as ill-posed under Hadamard's criteria, despite the solution's guaranteed existence and uniqueness. To address this challenge, we employ regularization methods that systematically suppress high-frequency noise while preserving spectral features. The methodology comprises two key components: the design of a regularization scheme to stabilize the inverse problem of the dispersion relation; and a mathematical optimization procedure to determine the desired physical solution. It should be emphasized that the inverse-problem-based framework reconstructs low-energy QCD dynamics ($\rho(s) \equiv \frac{1}{\pi} \text{Im}\Pi(s)$) directly from OPE inputs without phenomenological assumptions or tunable parameters, circumventing the quark-hadron duality assumption and Borel transformation endemic to traditional sum rules. Convergence is mathematically guaranteed: as OPE uncertainties vanish, the resulting solutions converge to the true physical observables [21].

This work pioneers the rigorous application of inverse problem approach to a kind of non-perturbative QCD quantity: the spectral properties of axial-vector current and the decay constants and of the pion and a_1 states. On the one hand, it is related to chiral symmetry, where low-energy constants (LECs) of chiral perturbation theory are directly expressible through spectral functions [22–24], enabling their theoretical determination to facilitate an in-depth understanding of non-perturbative QCD. On the other hand, the corresponding excited states are away from the threshold of a pair of light quarks, which offers an opportunity to probe the confinement mechanism. By implementing Tikhonov regularization with the quasi-optimality criterion [25], we stably reconstruct spectral functions – including excited states – directly from dispersion relations, circumventing the quark-hadron duality assumption and Borel transformation. Our approach reveals emergent resonance structures, and the non-trivial structure demonstrates that certain resonances must exist. The predicted decay constants for ground and excited states can be directly associated with leptonic decays, and these quantities are also important for determination of light-cone distribution amplitudes (LCDAs) [26, 27], which are pivotal in the study of hadronic decays [28–34]. Predictions such as $\mathcal{B}(\tau \rightarrow a_1(1260)\nu_\tau) = 0.184 \pm 0.032$ and $\mathcal{B}(\tau \rightarrow a_1(1640)\nu_\tau) = 0.015 \pm 0.004$ – testable at Belle II and future collider Super Tau-Charm Facility (STCF) – demonstrate the framework's predictive power. This methodology establishes a new paradigm for non-perturbative hadronic spectroscopy, with precision systematically improvable through higher-order OPE inputs. Besides, prediction for the spectral functions provides direct access to fundamental parameters of chiral perturbation theory in a model independent way [22–24, 35, 36].

The organization of this paper is as follows. Sec. 2 establishes the inverse-problem approach for determining QCD's low-energy description from dispersion relations, overcoming fundamental limitations of traditional sum rules. Sec. 3 explores our approach's predictive power, reconstructs spectra, extracts decay constants, and reveals their phenomenological implications. We finish with the conclusion of this work in Sec. 5.

2 Theoretical framework

In this section, we present the theoretical framework for extracting the pion decay constant within the inverse problem approach. The approach was originally introduced and systematically developed in Ref. [21]. Ref. [21] focuses primarily on the underlying mathematical formalism and employs three toy models to demonstrate the effectiveness of regularization methods in addressing the ill-posedness of inverse problems. In this work, we demonstrate the utility and potential of the inverse problem approach for non-perturbative QCD by applying it to the extraction of the pion decay constant.

We begin with the correlation function used to extract the pion decay constant, along with its dispersion relation and the corresponding operator product expansion (OPE) input. This setup follows the standard QCD sum rules framework. Subsequently, we formulate the dispersion relation as an inverse problem, using the OPE input to reconstruct non-perturbative quantities including the pion decay constant and the spectral function. This reconstruction marks the key departure from QCD sum rules. Finally, we briefly outline the Tikhonov regularization procedure to stabilize the ill-posedness of inverse problem.

2.1 Correlation function and Dispersion Relation

2.1.1 Correlation function

The decay constant of pion f_π is defined by the matrix element

$$\langle 0 | j_{\mu 5} | \pi \rangle \equiv i f_\pi p_\mu^\pi, \quad (2)$$

where $j_{\mu 5} = \bar{u} \gamma_\mu \gamma_5 d$ denotes the axial-vector current and p^π is the momentum of pion. As in QCD-based approaches, such as lattice QCD and QCD sum rules, the determination of such quantities begins with the construction of the appropriate correlation functions. Based on Eq. (2), we consider the following correlation function:

$$\Pi_{5\mu\nu}(q) = i \int d^4x e^{iq \cdot x} \langle 0 | T \{ j_{\mu 5}(x) j_{\nu 5}(0) \} | 0 \rangle = -g_{\mu\nu} \Pi_g(q^2) + q^\mu q^\nu \Pi_q(q^2), \quad (3)$$

with the QCD non-perturbative vacuum $|0\rangle$. In the deep Euclidean region, $q^2 \ll 0$, the correlation function is calculable within the OPE. By contrast, in the timelike region, $q^2 > 0$, the correlation function encodes the hadronic information. Inserting a complete set $\sum_n |n\rangle \langle n|$ of hadronic states that couple to $j_{\mu 5}$, we obtain

$$2\text{Im}\Pi_{5\mu\nu}(q) = \sum_n \langle 0 | j_{\mu 5} | n \rangle \langle n | j_{\nu 5} | 0 \rangle d\phi_n (2\pi)^4 \delta^4(q - p_n), \quad (4)$$

where $d\phi_n$ and p_n denote the phase space and the moment of state $|n\rangle$, respectively. For the axial-vector current $j_{\mu 5}$, the complete set $\{|n\rangle\}$ includes pseudoscalar states such as $|\pi\rangle$, axial-vector states such as $|a_1\rangle$, and continuum states such as $|3\pi\rangle$ and $|5\pi\rangle$. Consequently, the correlator in Eq. (4) encodes both the pion decay constant f_π defined in Eq. (2), and the decay constant f_{a_1} of the axial-vector state, defined as follows:

$$\langle 0 | j_{\mu 5} | a_1 \rangle \equiv \epsilon_\mu^{a_1} f_{a_1} M_{a_1}, \quad (5)$$

where $\epsilon_\mu^{a_1}$ and M_{a_1} are the polarization vector and mass of a_1 , respectively. The pion matrix element in Eq. (2) is proportional to the pion momentum p_μ , while the axial-vector matrix element in Eq. (5) contains the polarization vector ϵ_μ . Consequently, the invariant amplitude $\Pi_g(q^2)$ in Eq. (3) with structure $g_{\mu\nu}$ includes only the contribution a_1 , while $\Pi_q(q^2)$ receives contributions from both the pion and a_1 . To determine f_π and compare it with experiment, we therefore focus our analysis on the invariant amplitude $\Pi_q(q^2)$.

2.1.2 Dispersion Relation

Applying Cauchy's theorem to the contour introduced in Ref. [21], we obtain the dispersion relation for $\Pi_q(q^2)$,

$$\Pi_q(q^2) = \frac{1}{\pi} \int_{t_{\min}}^{\infty} ds \frac{\text{Im}\Pi_q(s)}{s - q^2} + \frac{1}{2\pi i} \oint_{|s|=R \rightarrow \infty} ds \frac{\Pi_q(s)}{s - q^2} \quad (6)$$

which relates the OPE representation of $\Pi_q(q^2)$ in the deep Euclidean region ($q^2 \ll 0$) to its spectral function in the physical timelike region ($q^2 > 0$). If $\Pi_q(s)$ vanishes as $|s| \rightarrow \infty$, the second term on the right-hand side of Eq. (6) vanishes, yielding the standard unsubtracted dispersion relation. For large $|s|$ away from the physical cut, the asymptotic behavior of $\Pi_q(s)$ is governed by its OPE. The OPE has been calculated through NLO in α_s and dimension six in the condensate expansion [37, 38]:

$$\begin{aligned} \Pi_q(q^2) = & -\frac{1}{4\pi^2} \left(1 + \frac{\alpha_s(\mu)}{\pi}\right) \ln \left(\frac{-q^2}{\mu^2}\right) + \frac{1}{12\pi} \frac{\langle \alpha_s G^2 \rangle}{(q^2)^2} \\ & + \frac{m_u \langle \bar{u}u \rangle + m_d \langle \bar{d}d \rangle}{(q^2)^2} + \frac{1}{9} \frac{m_u \langle g_s \bar{u} \sigma T G u \rangle + m_d \langle g_s \bar{d} \sigma T G d \rangle}{(q^2)^3} \\ & - \frac{4}{81} \frac{\langle g_s \bar{u}u \rangle^2 + \langle g_s \bar{d}d \rangle^2}{(q^2)^3} - \frac{\left[105 + 50 \ln \left(\frac{-q^2}{\mu^2}\right)\right]}{243\pi^2} \frac{\sum_{\psi=u,d,s} \langle g_s^2 \bar{\psi} \psi \rangle^2}{(q^2)^3}. \end{aligned} \quad (7)$$

The leading logarithm shows that $\Pi_q(s)$ does not vanish as $|s| = |-q^2| \rightarrow \infty$, but instead grows logarithmically. Therefore, the dispersion relation in Eq. (6) must be subtracted, for which the contour contribution at infinity vanishes. For a subtraction at $s = \sigma_0$, its dispersion relation is,

$$\Pi_q(\sigma_0) = \frac{1}{\pi} \int_{t_{\min}}^{\infty} ds \frac{\text{Im}\Pi_q(s)}{s - \sigma_0} + \frac{1}{2\pi i} \oint_{|s|=R \rightarrow \infty} ds \frac{\Pi_q(s)}{s - \sigma_0}, \quad (8)$$

Subtracting Eq. (8) from Eq. (6) yields

$$\frac{\Pi_q(q^2) - \Pi_q(\sigma_0)}{q^2 - \sigma_0} = \frac{1}{\pi} \int_{t_{\min}}^{\infty} ds \frac{\text{Im}\Pi_q(s)}{(s - q^2)(s - \sigma_0)} + \frac{1}{2\pi i} \oint_{|s|=R \rightarrow \infty} ds \frac{\Pi_q(s)}{(s - q^2)(s - \sigma_0)}, \quad (9)$$

where the last term vanishes for the logarithmical divergence $\Pi_q(s)$. We then obtain the once-subtracted dispersion relation,

$$\frac{\Pi_q(q^2) - \Pi_q(\sigma_0)}{q^2 - \sigma_0} = \frac{1}{\pi} \int_{t_{\min}}^{\infty} \frac{\text{Im}\Pi_q(s) ds}{(s - \sigma_0)(s - q^2)}, \quad (10)$$

where $\text{Im}\Pi_q(s)$ is determined from OPE for $s \gg 0$. Upon analytic continuation to $q^2 = s > 0$, the logarithms generate an imaginary part,

$$\ln \left(\frac{-s}{\mu^2}\right) = \ln \left(\frac{s}{\mu^2}\right) - \pi i, \quad (11)$$

yielding the reliable quark-hadron duality,

$$\text{Im}\Pi_q(s) = \text{Im}\Pi_q^{\text{pert}}(s) = \frac{1}{4\pi} \left(1 + \frac{\alpha(\mu)}{\pi}\right), \quad \text{for } s \gg 0. \quad (12)$$

2.2 Inverse Problem

2.2.1 Integral Equation

We split the dispersion integral at a separation scale Λ , identifying the regions below and above Λ as non-perturbative and perturbative, respectively. Then, the subtracted dispersion relation in Eq. (10) can

rewritten as an integral equation of inverse problem,

$$\int_{t_{\min}}^{\Lambda} \frac{\rho(s) ds}{(s - \sigma_0)(s - q^2)} = \frac{\Pi_q(q^2) - \Pi_q(\sigma_0)}{q^2 - \sigma_0} - \int_{\Lambda}^{\infty} \frac{\frac{1}{\pi} \text{Im}\Pi_q^{\text{pert}}(s) ds}{(s - \sigma_0)(s - q^2)}, \quad (13)$$

where $\rho(s) \equiv \text{Im}\Pi_q(s)/\pi$ denotes the spectral function. The spectral function $\rho(s)$ over $t_{\min} \leq s \leq \Lambda$, appearing on the left-hand side, is the non-perturbative quantity to be reconstructed. By contrast, all quantities on the right-hand side are theoretically calculable: $\Pi_q(q^2)$ and $\Pi_q(\sigma_0)$ are evaluated from the OPE in the deep Euclidean region, $q^2, \sigma_0 \ll 0$, while the quark-hadron duality is reliable for $s > \Lambda$. Therefore, the central task is to identify the low-energy non-perturbative spectral function from the known OPE input by solving this integral equation. This is a Fredholm integral equation of the first kind, which determine the unknown integrand from the known integral result and thus constitutes a typical inverse problem.

Before proceeding, we first discuss the admissible ranges of the parameters Λ and σ_0 . The separation scale Λ is not uniquely defined, since the transition between the perturbative and non-perturbative regimes is continuous rather than abrupt. Therefore, the physical results should be insensitive to variations of Λ within a reasonable range. The numerical analysis below confirms this expectation. Nevertheless, Λ cannot be chosen arbitrarily large or small. If Λ is too large, the reconstruction interval $[t_{\min}, \Lambda]$ becomes wider, increasing the amount of information to be determined and thereby reducing the reconstruction quality. Conversely, if Λ is too small, the perturbative expansion becomes less convergent, enhancing higher-order corrections and the neglected contributions, and thereby degrading the resulting reconstruction. We now turn to the determination of the subtraction point σ_0 . The subtraction at σ_0 improves the large- s behavior of the integrand, thereby ensuring convergence of the dispersion integral. However, the subtraction introduces $\Pi_q(\sigma_0)$ as an additional input. Its reliable evaluation requires σ_0 to lie sufficiently deep in the Euclidean region, where the OPE exhibits good convergence. Moreover, physical observables should be independent of the arbitrary choice of subtraction point. As shown below, the extracted results exhibit negligible dependence on σ_0 . In QCD sum rules, the subtraction point is typically chosen as $\sigma_0 = 0$. The subtraction constant $\Pi(0)$ vanishes identically in the vector channel, whereas in the axial-vector channel it receives non-perturbative contributions and is generally nonzero. Treating $\Pi_q(0)$ as an additional unknown would weaken the constraints on the inverse problem and thereby reduce the reliability of the extraction. The deep-Euclidean subtraction point adopted here avoids this complication.

2.2.2 Integral Equation with Prior Information

Eq. (13) provides the general form of the inverse problem integral equation derived from a once-subtracted dispersion relation. For the determination of f_π , the pion contribution to the spectral function takes the form of a delta-function pole at $s = m_\pi^2$. Directly reconstructing this delta-function contribution is, however, highly challenging. When the objective is to determine properties of the known pion state, such as its decay constant, LCDA, and PDF, rather than to establish the state itself, its measured mass m_π and delta-function pole structure can be incorporated as prior information. Incorporating this prior information, we decompose the spectral function into an isolated pion-pole contribution and a continuum contribution,

$$\rho(s) = f_\pi^2 \delta(s - m_\pi^2) + \rho^c(s) \theta(s - s_{\text{th}}), \quad (14)$$

where s_{th} denotes the physical hadronic threshold; for the correlator in Eq. (3), $s_{\text{th}} = (3m_\pi)^2$. The continuum spectral function $\rho^c(s)$ starts at s_{th} and may include contributions from excited states, as discussed in the numerical analysis below.

We emphasize that the explicit delta function in Eq. (14) serves only to make the physical interpretation transparent. This does not imply, however, that the delta function itself emerges from the mathematical inversion of the integral equation. Instead, for the actual mathematical implementation, we substitute

Eq. (14) into Eq. (13) and perform the integration, yielding

$$\frac{f_\pi^2}{(m_\pi^2 - \sigma_0)(m_\pi^2 - q^2)} + \int_{s_{\text{th}}}^\Lambda \frac{\rho^c(s) ds}{(s - \sigma_0)(s - q^2)} = \frac{\Pi_q(q^2) - \Pi_q(\sigma_0)}{(q^2 - \sigma_0)} - \int_\Lambda^\infty \frac{\frac{1}{\pi} \text{Im} \Pi_q^{\text{pert}}(s) ds}{(s - \sigma_0)(s - q^2)}. \quad (15)$$

We then need determine two distinct unknown components: the continuum spectral function $\rho^c(s)$ over $s_{\text{th}} \leq s \leq \Lambda$ and the isolated physical quantity f_π^2 . This mathematical problem is fundamentally different from that defined by Eq. (13).

In Ref. [21], we proved that the inverse problem of dispersion relation admits a unique solution but is unstable in the L^2 space. However, the δ distribution appearing in Eq. (14) does not belong to L^2 , since it is not square integrable. We therefore adopt the formulation in Eq. (15) rather than that in Eq. (13), thereby separating the pion-pole residue f_π^2 from the continuum spectral function $\rho^c(s)$. This reformulation changes the mathematical problem in principle, necessitating a new analysis of the properties of the integral equation. As demonstrated in the Appendix, the integral equation in Eq. (15) still admits a unique but unstable solution. It therefore remains ill-posed and must be solved using the regularization method.

2.2.3 Simplified the Integral Equation

In solving the integral equation, we choose $\sigma_0 \ll 0$. Since $s \geq s_{\text{th}} > 0$, the factor $s - \sigma_0$ remains strictly positive throughout the integration domain. We then introduce the redefinition,

$$\begin{cases} f^{\text{pole}} \equiv f_\pi^2 / (m_\pi^2 - \sigma_0), \\ f^c(s) \equiv \rho^c(s) / (s - \sigma_0), \\ g(q^2) \equiv \frac{\Pi_q(q^2) - \Pi_q(\sigma_0)}{(q^2 - \sigma_0)} - \int_\Lambda^\infty \frac{\frac{1}{\pi} \text{Im} \Pi_q^{\text{pert}}(s) ds}{(s - \sigma_0)(s - q^2)}. \end{cases} \quad (16)$$

Furthermore, Eq. (15) can be expressed,

$$\frac{f^{\text{pole}}}{m_\pi^2 - q^2} + \int_{s_{\text{th}}}^\Lambda ds \frac{f^c(s)}{s - q^2} = g(q^2) \quad (17)$$

where $q^2 \in [q_{\text{low}}^2, q_{\text{up}}^2]$ with $q_{\text{low}}^2 < q_{\text{up}}^2 \ll 0$, and $s \in [s_{\text{th}}, \Lambda]$ with $0 < m_\pi^2 < s_{\text{th}} < \Lambda$. Moreover, $f^c \in L^2(s_{\text{th}}, \Lambda)$ and $g \in L^2(q_{\text{low}}^2, q_{\text{up}}^2)$. Accordingly, the proof of ill-posedness presented in the Appendix is based on the form of Eq. (17).

2.2.4 Overview of the Inverse Problem Approach

As proven in the Appendix, Eq. (17) admits a unique but unstable solution and is therefore ill-posed, necessitating the use of regularization methods. The basic idea and a detailed introduction to regularization methods can be found in Ref. [21]. Regularization method is to replace the original ill-posed operator with a nearby well-posed approximation. Under Tikhonov regularization, the regularized approximate solution has been proven to converge to the true solution as the input error tends to zero. This convergence enables a rigorous analysis of statistical and systematic uncertainties, while allowing the precision to be systematically improved. These features will be demonstrated explicitly through the numerical results presented in the next chapter.

We now briefly outline the numerical procedure of the inverse problem approach. Following Ref. [21], we employ Tikhonov regularization. However, unlike the formulation considered therein, Eq. (17) contains two distinct contributions, a pole term and a continuum spectral function. We define the Tikhonov functional as $J_\alpha(f) = \|Kf - g^\delta\|_G^2 + \alpha \|f^c\|_F^2$, whose minimizer gives the regularized approximate solution, $f_\alpha^\delta = \arg \min_f J_\alpha(f)$. The first term in the functional is the residual term, which quantifies the agreement

between the reconstructed solution and the input data, analogous to that in a conventional χ^2 fit. The second one is the penalty term, with the regularization parameter α stabilizing the solution of the ill-posed problem against overfitting. For the problem considered here, the ill-posedness arises primarily from reconstructing the continuum spectral contribution; we therefore regularize only f^c through the penalty term $\alpha \|f^c\|_F^2$. It keeps f^c relatively smooth and suppresses spurious oscillations. We take $G = L^2$ as the input space and $F = H^1$ as the solution space for the continuum spectral function, thereby enforcing its smoothness. The space L^2 consists of square-integrable functions, whereas H^1 additionally requires their first weak derivatives to be square integrable. Detailed definitions are provided in the Appendix of Ref. [21]. We next discretize $f^c(s)$ using the finite-element method, $f^c(s) = \sum_i c_i \varphi_i(s)$ with the hat function $\varphi_i(s)$. For a linear problem, minimizing the functional J_α reduces to solving a linear system. Implementation details are provided in Sec. 4.4 of Ref. [21]. Finally, we obtain the regularized approximate solution f_α^δ , which depends on the regularization parameter α and the input error δ . Among the mathematical methods available for determining α , we adopt the quasi-optimality criterion, $\alpha_{\text{opt}} = \arg \min_\alpha \left\| \frac{\partial f_\alpha^c(s)}{\partial \ln \alpha} \right\|$. In practice, α is chosen within a plateau where f_α^δ is stable against variations in α , consistent with physical expectations. As a mathematical parameter, α should not significantly affect physical results, which are therefore expected to exhibit a stability plateau with respect to α . This plateau will be demonstrated numerically below.

As established in Ref. [21], under suitable a priori conditions, the Tikhonov regularization solution f_α^δ converges to the exact solution f as the input error $\delta \rightarrow 0$. This convergence is essential, as it ensures that the inverse problem approach can be systematically improved by reducing the input error. From another perspective, the inverse problem approach enables a numerical quantification of statistical and systematic uncertainties, which can be systematically reduced as the input error δ decreases. These uncertainties are also examined in detail in the numerical analysis below. Finally, for the underlying principles and a detailed discussion of regularization methods, see Ref. [21].

We now summarize the criteria for assessing the quality of a reconstructed solution. Specifically, a reliable solution must simultaneously exhibit stability plateaus with respect to both α and Λ . The α plateau reflects the requirement that physical results be insensitive to variations of this mathematical parameter. Furthermore, the Λ plateau indicates that the reconstructed spectral function approaches its perturbative counterpart near the upper boundary, $s \simeq \Lambda$.

3 Numerical Results

3.1 Results under Tikhonov Regularization

3.1.1 Input

The OPE calculation for the correlation function in the deep Euclidean region, given in Eq. (7), involves various vacuum condensates whose values are required for the numerical analysis. We adopt the values given in Refs. [37–39],

$$\begin{aligned}
\langle \alpha_s G^2 \rangle &= 0.07 \pm 0.02 \text{ [GeV}^4], \quad m_u \langle \bar{u}u \rangle + m_d \langle \bar{d}d \rangle = -(1.651 \pm 0.003) \times 10^{-4} \text{ [GeV}^4], \\
m_u \langle g_s \bar{u} \sigma T G u \rangle + m_d \langle g_s \bar{d} \sigma T G d \rangle &= -(1.321 \pm 0.033) \times 10^{-4} \left[\frac{\alpha_s(\mu)}{\alpha_s(2 \text{ GeV})} \right]^{14/(3\beta_0)} \text{ [GeV}^4], \\
\langle g_s \bar{u}u \rangle^2 = \langle g_s \bar{d}d \rangle^2 &= (2.082_{-0.697}^{+0.734}) \times 10^{-3} \left[\frac{\alpha_s(\mu)}{\alpha_s(2 \text{ GeV})} \right]^{4/(\beta_0)} \text{ [GeV}^6], \\
\sum_{\psi=u,d,s} \langle g_s^2 \bar{\psi} \psi \rangle^2 &= (2 + \kappa^2) \langle g_s^2 \bar{q}q \rangle^2, \quad \text{with } \langle g_s^2 \bar{q}q \rangle^2 = 7.420_{-2.483}^{+2.614} \times 10^{-3} \text{ [GeV}^6],
\end{aligned} \tag{18}$$

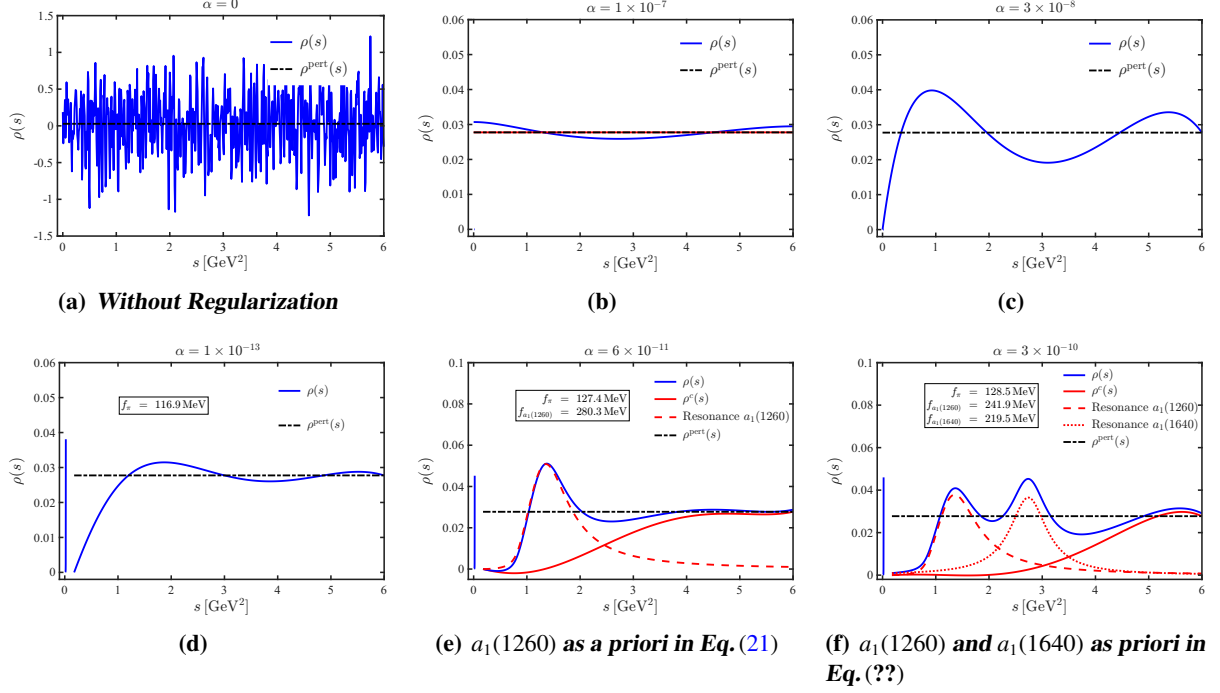


Figure 1: Numerical Results. (a) Without Regularization and (b-f) Under Tikhonov regularization; The subfigures (b) without any prior information; (c) with boundary condition; (d) with pion pole as a priori; (e) with pion pole and $a_1(1260)$ resonance; (f) with pion pole, $a_1(1260)$ and $a_1(1640)$ resonances.

where the scale dependence of the relevant condensates is included, $\beta_0 = 11 - \frac{2}{3}n_f$, and n_f denotes the number of the active quark flavors in the scale μ . Besides, the factor $\kappa \equiv \langle \bar{s}s \rangle / \langle \bar{q}q \rangle = 0.74 \pm 0.03$ ($q = u$ or d) [39]. It should be noted that most of the condensate values used here are extracted from QCD sum rules and therefore carry corresponding uncertainties. Although Eq. (18) quotes uncertainties for these condensates, their actual uncertainties may be larger. In the present analysis, we use these values as input. As more observables are calculated using the inverse problem approach, these condensates would be extracted.

3.1.2 Without Regularization

With these inputs, we numerically solve the integral equation in Eq. (15), or equivalently its simplified form in Eq. (17). We first examine the results obtained without regularization. As shown in Fig. 1(a), the reconstructed solution without regularization exhibits severe oscillations and yields no physically meaningful result. This behavior provides direct numerical evidence for the ill-posedness of the inverse problem. It further demonstrates that regularization method is indispensable for obtaining a stable solution to the inverse problem.

3.1.3 Tikhonov regularization

We now turn to present the numerical results obtained using Tikhonov regularization. In Sec. II, we derived two integral equations (13) and (15). We first solve the integral equation in Eq. (13) without incorporating prior information about the pion. The results are shown in Figs. 1(b) and 1(c), obtained without and with boundary conditions, respectively. As seen in Figs. 1(b) and 1(c), Tikhonov regularization effectively stabilizes the ill-posedness, yielding smooth and stable solutions contrast to the highly oscillatory result in Fig. 1(a).

The solution in Fig. 1(b) is obtained without any prior information or boundary conditions and is

therefore relatively trivial. Here, the perturbative solution is defined by $\int_0^\Lambda \frac{\frac{1}{\pi} \text{Im}\Pi^{\text{pert}}(s) ds}{s-q^2} = \Pi_q^{\text{pert}}(q^2) - \int_\Lambda^\infty \frac{\frac{1}{\pi} \text{Im}\Pi^{\text{pert}}(s) ds}{s-q^2}$, where $\Pi^{\text{pert}}(q^2)$ is evaluated with all condensate contributions excluded. The perturbative solution is obtained by evaluating the analytic expression for $\text{Im}\Pi^{\text{pert}}(s)$ over the interval $s \in [0, \Lambda]$. Specifically, the inverse-problem solution is obtained by using the right-hand side of this equation as input to reconstruct the integrand on the left-hand side. For comparison, the black dash-dotted line represents the perturbative solution, $\rho^{\text{pert}}(s) \equiv \text{Im}\Pi^{\text{pert}}(s)/\pi$, while the red solid line shows the inverse problem solution obtained from the purely perturbative input. As shown, the two curves nearly coincide, demonstrating that the inverse problem approach successfully reproduces the perturbative solution. This agreement thereby confirms the ability of the inverse-problem approach to recover the desired solution. We then include all condensate contributions in the input, yielding the result shown by the blue solid line. Consequently, the difference between the blue and red curves reflects the impact of the condensate contributions. Here, the condensate contributions are much small than the perturbative term, resulting in only a small difference between the two solutions. The difference between the blue and red curves is also consistent with the uniqueness of the inverse problem solution, as distinct inputs lead to distinct reconstructed solutions.

In Fig. 1(c), we impose $\rho(s_{\text{th}}) = 0$ at threshold and match the spectral function at $s = \Lambda$ to the imaginary part of the full OPE, as implied by quark-hadron duality at high energies. The marked difference from Fig. 1(b) highlights the importance of imposing appropriate boundary conditions. The pronounced peak in the spectral function signals the presence of hadronic states. Since appropriate boundary conditions are a natural requirement of any physical problem, the solution in Fig. 1(c) provides a more meaningful physical description.

The broad structure in Fig. 1(c) lacks sufficient resolution to isolate the pion pole. We therefore incorporate the known pion state as prior information, which corresponds precisely to the inverse problem in Eq. (15). Incorporating the pion pole with only its mass specified, we extract f_π from the inverse problem, as shown in Fig. 1(d). For the continuum spectral function $\rho^c(s)$, the lower boundary is set at the three-pion threshold, $s_{\text{th}} = (3m_\pi)^2$. The extracted value, $f_\pi = 116.9$ MeV, is already in reasonable agreement with the experimental value $f_\pi^{\text{exp}} = 130.4$ MeV. This agreement demonstrates the effectiveness of the inverse problem approach for determining f_π . We now turn to the continuum spectral function, which exhibits a broad enhancement, indicating additional hadronic structure beyond the pion and sizable contributions from higher hadronic states. However, the limited resolution prevents the extraction of detailed information about the underlying hadronic states. The limited resolution has two main sources. First, the OPE input is currently known only through next-to-leading order in α_s , with condensate contributions included up to dimension six. Second, Tikhonov regularization has limited resolving power for Breit-Wigner peaks, as discussed in Ref. [21]. Further improvements therefore require progress on both fronts. In addition, the reconstruction can be further improved by incorporating prior information.

In addition, the reconstruction can be further improved by incorporating prior information about the resonance states. To this end, we note that the axial-vector current $j_{\mu 5}$ couples to both pseudoscalar and axial-vector mesons. Experimentally, several mesons with $J^P = 0^-$ and 1^+ have been observed. The contributions of pseudoscalar pion excitations π' to $\text{Im}\Pi_q(s)$ are suppressed by the light current-quark masses $m_{u,d}$ [40–42] and can therefore be neglected. By contrast, in the axial-vector channel, the PDG [43] lists states such as the $a_1(1260)$ and $a_1(1640)$, whose contributions we examine below. Unlike the pion pole, which is represented by a delta function, the a_1 resonances have finite widths and are described by relativistic Breit-Wigner forms. The Breit-Wigner distribution for the $a_1(1260)$ is given by,

$$\text{BW}(s, M_{a_1}, \Gamma_{a_1}) = \frac{1}{\pi} \frac{M_{a_1} \Gamma_{a_1}(s)}{(s - M_{a_1}^2)^2 + (M_{a_1} \Gamma_{a_1}(s))^2}, \quad (19)$$

where the energy-dependent width is defined as $\Gamma_{a_1}(s) = \Gamma_{a_1} h(s)/h(M_{a_1}^2)$, with $h(s)$ denoting the phase-space factor [43]. Since the three-body decay $a_1 \rightarrow 3\pi$ involves a complicated phase-space structure and

is dominated by the $a_1 \rightarrow \rho\pi$ channel, we retain only the $\rho\pi$ contribution [44], for which

$$h(s) = \int_{4M_\pi^2}^{(\sqrt{s}-M_\pi)^2} \frac{ds'}{\pi} \frac{M_\rho \Gamma_\rho}{(s' - M_\rho^2)^2 + (M_\rho \Gamma_\rho)^2} \frac{\sqrt{\lambda(s, s', M_\pi^2)}}{16\pi s}, \quad (20)$$

with the Källén function $\lambda(x, y, z) = x^2 + y^2 + z^2 - 2xy - 2yz - 2xz$. Finally, the spectral function including the pion and $a_1(1260)$ contributions is written as,

$$\rho(s) = f_\pi^2 \delta(s - M_\pi^2) + f_{a_1(1260)}^2 \mathbb{B}\mathbb{W}(s, M_{a_1(1260)}, \Gamma_{a_1(1260)}) + \rho^c(s). \quad (21)$$

Here, the prior information consists of m_π , $m_{a_1(1260)}$, and $\Gamma_{a_1(1260)}$, while f_π , $f_{a_1(1260)}$, and $\rho^c(s)$ are determined from the inverse problem approach. The PDG gives $M_{a_1(1260)} = 1202 \pm 15$ MeV and $\Gamma_{a_1(1260)} = 422 \pm 12$ MeV. Both quantities were previously subject to large uncertainties, particularly the width. Their precision has improved substantially, largely owing to the high-precision determination of the $a_1(1260)$ resonance parameters from the LHCb analysis of $D^0 \rightarrow K^\mp \pi^\pm \pi^\pm \pi^\mp$ decays [45]. Accordingly, we adopt these two values in our analysis, as given by the PDG averages. To account for the possible overlap between the continuum spectrum and excited resonances, we retain the threshold $s_{\text{th}} = (3m_\pi)^2$ for $\rho^c(s)$. We show the final result in Fig. 1(e), where the red solid line denotes the continuum contribution, the red dashed line denotes the $a_1(1260)$ resonance, and the blue solid line represents the total spectral function. As is clearly seen, the $a_1(1260)$ peak demonstrates that this resonance makes a significant contribution to the spectral function. More importantly, after the $a_1(1260)$ resonance contribution is separated, the reconstructed continuum $\rho^c(s)$ remains close to zero at low s and begins to rise gradually beyond the $a_1(1260)$ resonance region. For $s \gtrsim 4$ GeV², the continuum approaches a plateau and agrees closely with the perturbative result. Taken together, this behavior confirms the sizable contribution of the $a_1(1260)$ and supports the physical consistency of the reconstructed continuum. Moreover, no discernible contributions from additional resonances are observed in the reconstructed continuum.

Since it remains unclear whether additional resonances contribute to the spectral function, we further examine solutions including multiple a_1 states. The PDG also lists the $a_1(1640)$; we therefore include this state and show the resulting spectral function in Fig. 1(f). We then compare Figs. 1(e) and 1(f). First, an additional $a_1(1640)$ peak appears, but it is lower than the $a_1(1260)$ peak, indicating a smaller contribution to the spectral function. Second, after including the $a_1(1640)$, the continuum near the upper endpoint shows poorer agreement with the perturbative result than that in Fig. 1(e). As discussed in the preceding section, the transition between the perturbative and non-perturbative regimes is smooth, and the extracted results should therefore remain insensitive to Λ over a reasonable range. Moreover, quark-hadron duality is expected to hold above Λ , so the reconstructed spectral function should approach the perturbative result toward the upper end of the solution interval. This behavior provides an important criterion for assessing the reliability of the reconstructed solution. Indeed, the continuum in Fig. 1(e) is nearly flat for $s \gtrsim 4$ GeV² and agrees closely with the perturbative result, whereas that in Fig. 1(f) exhibits fluctuation over $4 \lesssim s \lesssim 6$ GeV². We therefore favor the scenario in Fig. 1(e), which incorporates prior information only for the pion pole and the $a_1(1260)$ resonance, without introducing additional resonance states. At the current level of input precision, we expect the contributions from higher excited states to be contained in the continuum, as they cannot yet be resolved separately. Resolving additional excited states will require more precise input and is left for future investigation.

We now address several additional points. First, we discuss the negative continuum solution at small s in Fig. 1(e). Physically, the spectral representation of the correlation function in Eq. (4) implies $\rho(s) \propto |\langle 0 | j_{\mu 5} | n \rangle|^2$, which is positive semidefinite and therefore nonnegative. Nevertheless, the reconstructed continuum in Fig. 1(e) exhibits a small negative region. The small negative region arises because positivity is not explicitly imposed in solving the inverse problem in Eq. (13). For the present analysis, the negative excursion in Fig. 1(e) is small and does not affect our discussion or conclusions. We have therefore not pursued a dedicated implementation of the nonnegativity constraint in the numerical algorithm. With more precise input in future analyses, we expect the negative-solution issue to be resolved naturally. Second,

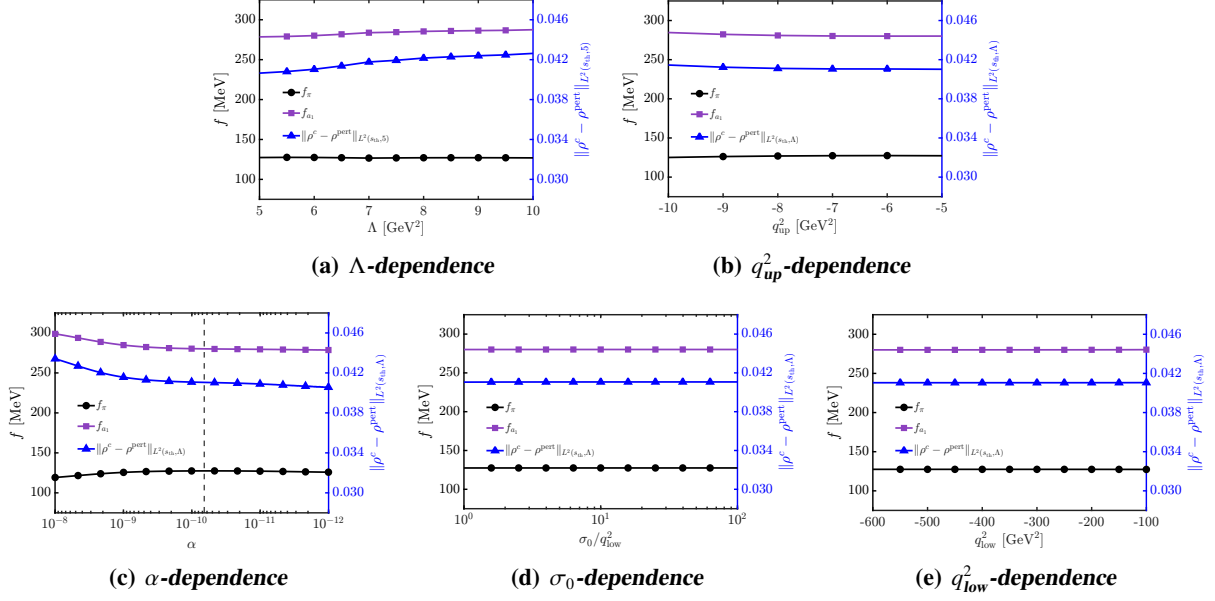


Figure 2: Our predicted values for the pion and $a_1(1260)$ decay constant exhibit minimal sensitivity to the choice of the cutoff Λ , subtraction point σ_0 , as well as q^2 interval.

we discuss the incorporation of prior information about the pion and a_1 mesons. As discussed in Ref. [21], the solution of an inverse problem generally improves as more reliable prior information is incorporated. Accordingly, adding prior information can improve the accuracy of the reconstructed solution. Conversely, can these peak structures be reconstructed directly through the mathematical inversion? We argue that they can. That this has not yet been achieved is due to the intrinsic smoothing effect of Tikhonov regularization, which tends to suppress peak structures. We expect that exploring alternative regularization methods in future work may allow the peak structures to be reconstructed directly. In the present work, however, we retain Tikhonov regularization and incorporate the corresponding prior information. Finally, we discuss the impact of the specific form of the integral kernel in the dispersion relation on the numerical results. Equation (15) is the original integral equation, which is transformed into the form of Eq. (17) through the redefinition in Eq. (16). The integral kernel in Eq. (17) is $1/(s - q^2)$. We adopt this form because it coincides with that studied in Ref. [21], where the relevant mathematical properties were established. Other integral kernels, such as $(q^2 - \sigma_0)/[(s - q^2)(s - \sigma_0)]$ and $(q^2 - \sigma_0)/(s - q^2)$, can also be employed. The integral equations associated with these three kernels are equivalent. To verify this equivalence numerically, we also solve the inverse problem using all three kernels. The extracted values of f_π differ by only 1 MeV, while those of f_{a_1} differ by less than 3 MeV, confirming the numerical equivalence of the three formulations.

3.2 Uncertainty Analysis

3.2.1 Parameter Stability

We now turn to an analysis of the uncertainties in the reconstructed solution. We first assess the robustness of the method by examining the sensitivity of the reconstructed solution to variations of the relevant parameters. Insensitivity to these parameters provides a key criterion for assessing the reliability of the inverse problem solution. We show the dependence of the reconstructed solution on the various parameters in Fig. 2.

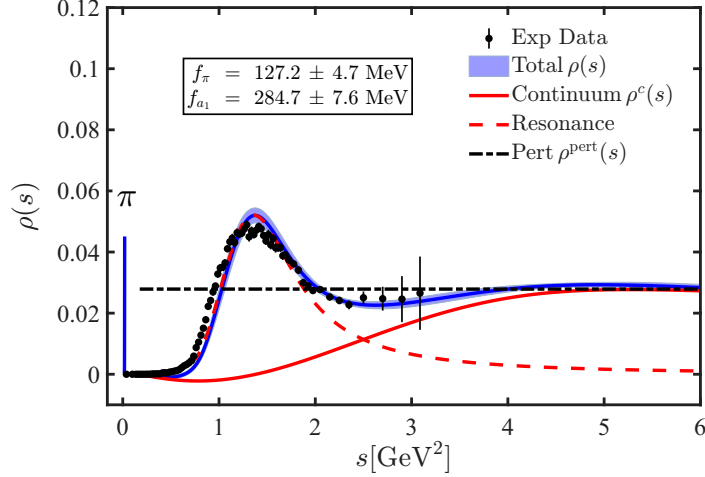


Figure 3: The spectral function are calculated by inverse problem approach.

3.2.2 Statistical and Systematic Uncertainties

The numerical results obtained above exhibit stability plateaus with respect to all parameters, supporting the reliability of the method. On this basis, we can now perform a detailed uncertainty analysis. As shown in Ref. [21], the uncertainty is quantified by the difference between the regularized approximate solution f_α^δ and the true solution f , and is divided by two contributions,

$$\|f_\alpha^\delta - f\| \leq \delta \|R_\alpha\| + \|R_\alpha K f - f\|. \quad (22)$$

The first term quantifies the uncertainty arising from the input error δ and is identified as the statistical uncertainty. To obtain the statistical uncertainty, we scan all input parameters including the vacuum condensates and the $a_1(1260)$ mass and width entering as prior information. The second term describes the approximation uncertainty introduced by replacing the exact inverse with the regularized operator R_α . The resulting deviation of the reconstructed solution is identified as the systematic uncertainty inherent in the method. To estimate this systematic uncertainty in practice, we replace the unknown true solution f with the numerical solution f_α^δ . A third source of uncertainty arises from parameter variations. As shown in Fig. 2, all parameters exhibit stability plateaus; their associated uncertainties are therefore estimated by varying them within the corresponding plateau regions. These contributions are included in the total systematic uncertainty. The final results, together with their estimated uncertainties, are shown in Fig. 3. The red solid line denotes the continuum contribution $\rho^c(s)$, the red dashed line the resonance contribution, and the blue solid line the total spectral function $\rho(s)$, while the shaded band represents the corresponding uncertainty of $\rho(s)$.

3.2.3 Reliability of the Results

We assess the reliability of our results by confronting them with experimental data in three respects. First, we compare the f_π with the experimental value and previous determinations from lattice QCD and QCD sum rules in Table 1. Our result, $f_\pi = 127.2 \pm 4.7$ MeV, agrees with the experimental value $f_\pi^{\text{exp}} = 130.57 \pm 0.14$ MeV within uncertainties. It is also consistent with previous determinations from lattice QCD and QCD sum rules. At the present level of input precision, our uncertainty remains larger than those from experiment and lattice QCD. The precision of the extracted f_π alone therefore does not yet showcase the full potential of the inverse problem approach.

Instead, the principal advantage of the inverse problem approach is its simultaneous access to excited states and the continuum spectral function. We next turn to the full continuum spectral function, which can be compared directly with experimental data. The black points with error bars in Fig. 3 represent the

Table 1: Decay constants.

Method	f_π [MeV]	$f_{a_1(1260)}$ [MeV]
This work	127.2 ± 4.7	284.7 ± 7.6
Experiment	130.57 ± 0.14 [43]	
Lattice QCD	130.20 ± 0.80 [51]	240.0 ± 28.5 [52]
QCDSR(Π_q)	125 [13], 135 ± 1 [42]	151 ± 20 [42]
QCDSR(Π_g)		238 ± 10 [27]

Note: In lattice studies [52], the reported values $f_{a_1(1260)} = 0.30 \pm 0.03$ [GeV²] with dimension GeV², which correspond to the product $f_{a_1} M_{a_1}$ in our notation. To facilitate comparison, we have converted their results by dividing by the respective a_1 masses quoted in their publications.

experimental spectral data extracted from τ -decay measurements by the ALEPH Collaboration [46–49]. These data correspond to the axial-vector channel and include the sum over all relevant hadronic final states, such as 3π , 5π , and πKK . As shown in Fig. 3, our numerical result is in excellent agreement with the experimental data. This agreement provides one of the most compelling validations of the inverse problem approach. From the threshold $s = (3m_\pi)^2$ up to $s = 3 \text{ GeV}^2$, the reconstructed spectrum agrees with the experimental data over nearly the entire range. In the region $(3m_\pi)^2 \leq s \lesssim 2 \text{ GeV}^2$, the spectrum is dominated by the $a_1(1260)$. Its peak position and width are determined by the prior information, whereas its height is determined by the extracted f_{a_1} ; the agreement with the experimental data therefore indicates that f_{a_1} is well determined. The spectrum is slightly shifted toward larger s relative to the experimental data, because the input $a_1(1260)$ width includes only the $\rho\pi$ contribution, whereas the actual experimental line shape involves more complicated decay dynamics [50]. Moreover, the PDG input is based primarily on measurements of the $D \rightarrow K\pi\pi\pi$ decay [45], which may introduce an additional error.

At higher energies, the agreement with experiment over $2 \text{ GeV}^2 \lesssim s \lesssim 3 \text{ GeV}^2$ validates the accuracy of the reconstructed continuum spectral function $\rho^c(s)$. Besides, despite the sizable experimental uncertainties near $s = 3 \text{ GeV}^2$, our result remains in close agreement with the experimental central values. Over $4 \text{ GeV}^2 \lesssim s \lesssim 6 \text{ GeV}^2$, the reconstructed continuum $\rho^c(s)$ agrees closely with the perturbative spectral function $\rho^{\text{pert}}(s)$, providing further validation of the theoretical framework and supporting the applicability of quark-hadron duality in this region. Near $s = 0.5 \text{ GeV}^2$, however, our result lies slightly below the experimental data because the reconstructed $\rho^c(s)$ becomes negative in this region. This artifact can be addressed in future analyses by improving the precision of the theoretical input or imposing an explicit nonnegativity constraint.

Finally, we discuss the decay constant of the $a_1(1260)$. Table 1 compares our result with determinations from lattice QCD and QCD sum rules. Our value is larger than the previous determinations from both approaches. The magnitude of $f_{a_1(1260)}$ can also be assessed from the spectral function in Fig. 3, where our result agrees well with the experimental data. With the $a_1(1260)$ mass and width fixed, $f_{a_1(1260)}$ controls the peak height; the successful reproduction of the experimental peak therefore supports our extracted value. We emphasize that $f_{a_1(1260)}$ is obtained solely from the inverse problem approach; the experimental data are not used as fitting input. Its agreement with the experimental data therefore provides an independent validation of the reliability of our result. The lattice QCD determination dates back to 1995 [52], and its uncertainty should be viewed in light of the computational limitations of that time. Two QCD sum rules determinations of f_{a_1} were reported in 2007 [27] and 2015 [42]. Both studies treated the $a_1(1260)$ in the zero-width approximation, representing its spectral contribution by a delta function. The 2007 analysis extracted f_{a_1} from the invariant amplitude Π_g in Eq. (3), where the $a_1(1260)$ is the lowest-lying state, making its determination relatively clean and less susceptible to systematic bias. By contrast, the 2015 analysis used Π_q , which receives contributions from both the pion and the $a_1(1260)$, rendering their simultaneous determination more challenging. Such a simultaneous treatment of multiple states is precisely a distinctive strength of the inverse problem approach.

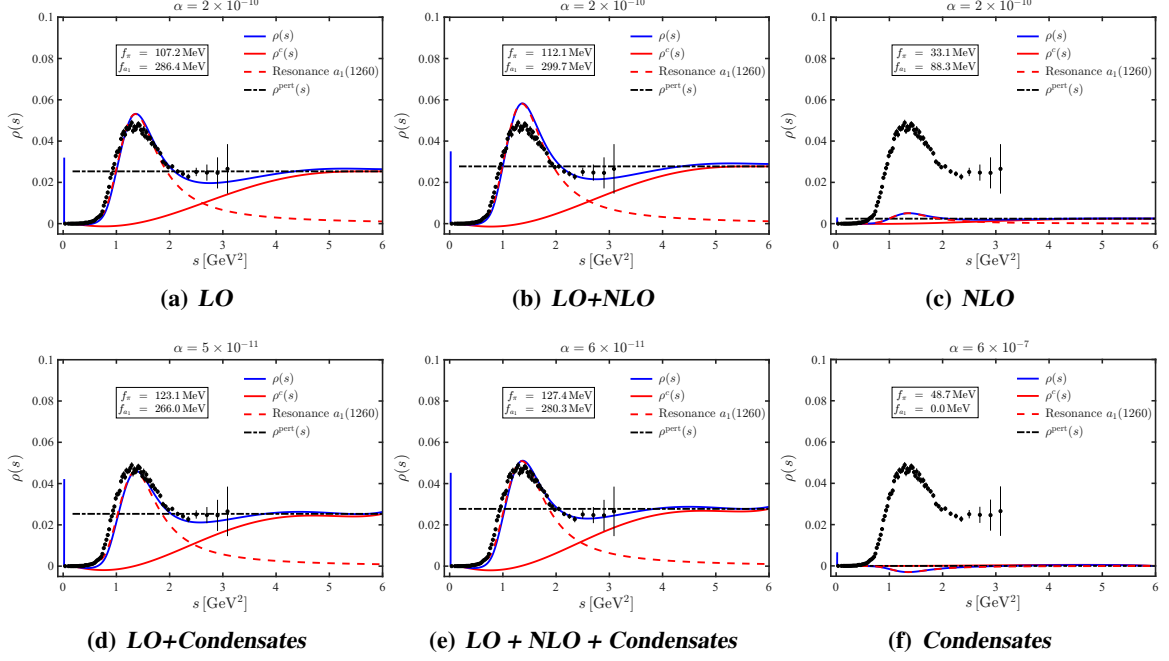


Figure 4: Convergence.

3.3 Convergence of Solutions and Precision Improvement

A crucial property of the inverse problem approach is that the approximate solution converges to the true solution as the input error tends to zero. A rigorous mathematical proof of this convergence was given in Ref. [21]. Beyond the idealized limit $\delta \rightarrow 0$, the toy model analyzed in Ref. [21] demonstrates that the solution error decreases systematically as the input precision improves. In this subsection, we examine the convergence of the solution obtained with the inverse problem approach for the specific physical problem considered here.

We first specify the pion pole and the $a_1(1260)$ resonance as prior information. To examine convergence, we consider four levels of theoretical input: LO, LO+NLO, LO+condensates, and LO+NLO+condensates. The corresponding results are shown in Fig. 4. With successive improvements to the theoretical input, the extracted result moves systematically toward the experimental value, exhibiting a clear convergence. This example thus demonstrates the systematic improvability of the inverse problem approach, with increased input precision yielding an increasingly accurate reconstruction.

In addition, Fig. 4 shows the results obtained using the NLO correction and the condensate contributions separately as input. In both cases, however, the resulting reconstruction is unsatisfactory. This is physically natural, since a meaningful solution requires the input to capture the dominant dynamics. Indeed, the LO input alone already yields a reasonably well solution.

4 Physical Discussions

4.1 $\Pi_g(q^2)$ v.s. $\Pi_q(q^2)$

In the preceding section, we applied the inverse problem approach to invariant amplitude $\Pi_q(q^2)$ to determine f_π , $f_{a_1(1260)}$, and the continuum spectral function. Besides Π_q , the correlation function in Eq. (3) contains the invariant amplitude Π_g , which couples to the a_1 states but not to the pion. Therefore, the a_1 decay constant can also be extracted from $\Pi_g(q^2)$. Does it agree with the value obtained from $\Pi_q(q^2)$? In this subsection, we turn to extract $f_{a_1(1260)}$ from the invariant amplitude Π_g .

Table 2: Individual contributions to the uncertainties of f_π and $f_{a_1(1260)}$. For each entry, only the indicated input is varied while all other inputs are fixed at their central values.

Source	Input	Sampling prescription	Δf_π [MeV]	Δf_{a_1} [MeV]
<i>Statistical</i>	$\langle \alpha_s GG \rangle$	$0.07 \pm 0.02 \text{ GeV}^4$	3.95	4.98
	μ	$[1, 3] \text{ GeV}$	2.16	0.46
	$\langle g_s^2 \bar{q}q \rangle^2$	$7.420_{-2.483}^{+2.614} \times 10^{-3} [\text{GeV}^6]$	1.07	1.64
	$M_{a_1(1260)}$	$1202 \pm 15 \text{ MeV}$	0.53	3.70
	$\langle g_s \bar{q}q \rangle^2$	$(2.082_{-0.697}^{+0.734}) \times 10^{-3} [\text{GeV}^6]$	0.42	0.68
	$\Gamma_{a_1(1260)}$	$422 \pm 12 \text{ MeV}$	0.08	2.17
	κ	0.74 ± 0.03	0.05	0.12
	$m_u \langle \bar{u}u \rangle + m_d \langle \bar{d}d \rangle$	$-(1.651 \pm 0.003) \times 10^{-4} [\text{GeV}^4]$	0.01	0.09
	$m_u \langle g_s \bar{u}\sigma TGu \rangle + m_d \langle g_s \bar{d}\sigma TGd \rangle$	$-(1.321 \pm 0.033) \times 10^{-4} [\text{GeV}^4]$	0.01	0.08
	Total statistical uncertainty		4.68	6.83
<i>Systematic</i>	Λ	$[5, 10] \text{ GeV}^2$	0.66	3.34
	q_{up}^2	$[-10, -5] \text{ GeV}^2$	0.30	0.51
	q_{low}^2	$[-600, -200 \text{ GeV}^2]$	0.03	0.10
	$\sigma_0/q_{\text{low}}^2$	$[1, 100]$	0.01	0.09
	α	2×10^{-13}	0.05	0.50
	Total systematic uncertainty		0.73	3.42
Total uncertainty			4.73	7.63

The Lorentz decomposition in Eq. (3) shows that the mass dimension of $\Pi_g(q^2)$ exceeds that of $\Pi_q(q^2)$ by two. In the deep Euclidean region, $q^2 \ll 0$, the corresponding OPE input is given by,

$$\begin{aligned}
\Pi_g(q^2) = & -\frac{1}{4\pi^2} \left(1 + \frac{\alpha_s(\mu)}{\pi}\right) q^2 \ln \left(\frac{-q^2}{\mu^2}\right) + \frac{1}{12\pi} \frac{\langle \alpha_s G^2 \rangle}{(q^2)} \\
& - \frac{m_u \langle \bar{u}u \rangle + m_d \langle \bar{d}d \rangle}{(q^2)} + \frac{1}{9} \frac{m_u \langle g_s \bar{u}\sigma TGu \rangle + m_d \langle g_s \bar{d}\sigma TGd \rangle}{(q^2)^2} \\
& - \frac{4}{81} \frac{\langle g_s \bar{u}u \rangle^2 + \langle g_s \bar{d}d \rangle^2}{(q^2)^2} - \frac{\left[105 + 50 \ln \left(\frac{-q^2}{\mu^2}\right)\right]}{243\pi^2} \frac{\sum_{\psi=u,d,s} \langle g_s^2 \bar{\psi}\psi \rangle^2}{(q^2)^2},
\end{aligned} \tag{23}$$

This expression is obtained from the following relation between Π_g and Π_q ,

$$\Pi_{(g)}(q^2) - q^2 \Pi_{(q)}(q^2) = -\frac{2(m_u + m_d) \langle \bar{q}q \rangle}{q^2} + \mathcal{O}((m_u + m_d)^2). \tag{24}$$

Moreover, the Gell-Mann–Oakes–Renner relation,

$$f_\pi^2 m_\pi^2 = -2(m_u + m_d) \langle \bar{q}q \rangle + \mathcal{O}(m_q^2), \tag{25}$$

shows that the difference between the corresponding Π_g and Π_q expressions is precisely the pion contribution.

As $|q^2| \rightarrow \infty$, $\Pi_g(q^2)$ grows as q^2 , up to logarithmic corrections, and therefore requires a twice-subtracted dispersion relation of the form,

$$\int_{s_{\text{th}}}^\Lambda \frac{\rho^{\Pi_g}(s) ds}{(s - \sigma_0)^2 (s - q^2)} = \frac{\Pi_g(q^2) - \Pi_g(\sigma_0) - (q^2 - \sigma_0) \Pi_g'(\sigma_0)}{(q^2 - \sigma_0)^2} - \int_\Lambda^\infty \frac{\frac{1}{\pi} \text{Im} \Pi_g^{\text{pert}}(s)}{(s - \sigma_0)^2 (s - q^2)} ds. \tag{26}$$

All parameters are chosen identically to those in the preceding Π_q analysis, and the resulting numerical solution is shown in Fig. 5. First, the f_{a_1} value extracted from the invariant amplitude $\Pi_g(q^2)$ agrees

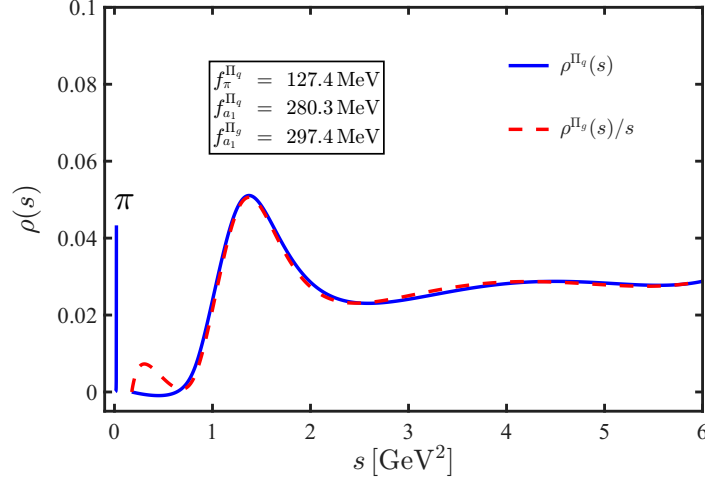


Figure 5: $\Pi_g(q^2)$ v.s. $\Pi_q(q^2)$

closely with that obtained from $\Pi_q(q^2)$. Turning to the spectral functions, $\rho^{\Pi_g}(s)/s$ and $\rho^{\Pi_q}(s)$ nearly coincide at large s and remain in close agreement even in the low- s region. The agreement demonstrates the internal consistency of the inverse problem approach and supports its unified description of the spectral dynamics.

4.2 Quark-Hadron Duality

4.2.1 Parameter Λ and Quark-Hadron Duality

As discussed in the preceding section, the continuum spectral function closes its perturbative counterpart near the upper endpoint, $s \approx \Lambda$. This matching is a manifestation of quark-hadron duality: the OPE provides a quark-level description of the correlation function, whereas the reconstructed spectral function encodes the corresponding hadronic dynamics. In Eq. (13), quark-hadron duality is imposed only for $s \geq \Lambda$, where the spectral function entering the integral over $[\Lambda, \infty)$ is identified with its perturbative counterpart. By contrast, throughout $0 \leq s \leq \Lambda$, the spectral function is determined by the inverse problem approach itself. Its automatic close to the perturbative result as $s \lesssim \Lambda$ reveals the onset of quark-hadron duality within the reconstructed region. The smooth matching across Λ provides a self-consistent description of the transition between the hadronic and perturbative regimes. Numerically, our results indicate that quark-hadron duality provides a valid description for $s \gtrsim 5 \text{ GeV}^2$.

In the inverse problem approach, Λ is introduced to separate the perturbative and non-perturbative regimes. Since this transition is continuous, Λ should be interpreted as lying in the region where quark-hadron duality sets in, rather than as marking a unique boundary point. Across this region, the solution is insensitive to variations of Λ and consequently exhibits a stability plateau, while quark-hadron duality is realized on both sides of Λ . Crucially, Λ is conceptually distinct from and should not be identified with, the effective continuum threshold s_0 employed in QCD sum rules.

In QCD sum rules, the hadronic spectral function is generally parametrized by a ground-state pole represented by a delta function and a continuum introduced through a theta-function,

$$\rho(s) = f_H^2 \delta(s - m_H^2) + \rho^c(s) \theta(s - s_{\text{th}}^H), \quad (27)$$

which is similar to Eq. (14). However, the continuum spectrum is not determined within the sum rules. Instead, the quark-hadron duality is imposed immediately above the hadronic threshold s_{th}^H . Specifically, for an unsubtracted dispersion relation, the global quark-hadron duality ansatz gives,

$$\int_{s_{\text{th}}^H}^{\infty} ds \frac{\rho^c(s)}{s - q^2} \simeq \int_{s_0}^{\infty} ds \frac{\frac{1}{\pi} \text{Im} \Pi^{\text{pert}}(s)}{s - q^2}. \quad (28)$$

Consequently, the hadronic spectral integral from the physical threshold s_{th}^H to infinity is replaced by the perturbative integral over $s \in [s_0, \infty)$. Then, the only remaining unknowns in the sum rule are then the ground-state residue f_H^2 and mass m_H^2 when the s_0 is given.

However, the physical threshold s_{th}^H is typically small and may even be lower than the ground-state pole m_H^2 [18]. The corresponding region is intrinsically non-perturbative, where a perturbative description is not reliable. Since the hadronic and perturbative spectral functions differ near the non-perturbative region, $s_0 \neq s_{\text{th}}^H$ is required to match their integrals and implement global quark-hadron duality. As a consequence, s_0 becomes a process-dependent phenomenological parameter that cannot be determined from first principles, and the extracted results are often sensitive to its value. In some physical applications, the uncertainty associated with s_0 constitutes the dominant source of error in QCD sum rules predictions.

We now turn to a comparison between the effective threshold s_0 in QCD sum rules and the separation scale Λ in the inverse problem approach. Formally, both serve as the lower limits of the spectral integrals over which quark-hadron duality is imposed. In practice, however, s_0 typically lies in the low-energy non-perturbative region, whereas Λ is chosen in a regime where perturbation theory is reliable. More specifically, in QCD sum rules analyses based on $\Pi_q(q^2)$, one finds $s_0^q \simeq 0.7 \text{ GeV}^2$ when f_π is extracted without an explicit a_1 contribution [18], whereas including the a_1 raises the effective threshold to $s_0 \simeq (1.3\text{--}1.7) \text{ GeV}^2$ [42]. By contrast, the extraction of f_{a_1} from $\Pi_g(q^2)$ employs $s_0 = 2.55 \pm 0.15 \text{ GeV}^2$ [27]. These examples show that s_0 depends not only on the physical quantity being extracted but also on the choice of invariant amplitude within the same correlation function. Consequently, s_0 must generally be determined separately for each analysis, introducing an empirical element and a significant sensitivity of the extracted results to its chosen value. The underlying reason is that QCD sum rules employ quark-hadron duality in the low-energy region, thereby ignoring the detailed excited-state and continuum contributions to calculate the ground-state properties. Moreover, since local quark-hadron duality $\rho(s) = \frac{1}{\pi} \text{Im} \Pi^{\text{pert}}(s)$ is clearly invalid in the low-energy region, it must instead rely on global quark-hadron duality.

The inverse problem approach enables the simultaneous reconstruction of the entire non-perturbative spectrum, including the ground state, excited states, and the continuum. The ability to reconstruct the continuum contribution constitutes the essential distinction from QCD sum rules. Accordingly, quark-hadron duality is invoked only in the region where perturbation theory is reliable. More specifically, we take $\Lambda \in [5, 10] \text{ GeV}^2$, substantially larger than the values of s_0 typically adopted in QCD sum rules analyses. Besides, extracting f_{a_1} from different invariant amplitudes (Π_q, Π_g) with the same Λ yields nearly identical results, and all of which exhibit negligible sensitivity to variations of Λ . In contrast to QCD sum rules, the inverse problem approach employs local quark-hadron duality for $s \gtrsim \Lambda$, where the hadronic and perturbative spectral functions are matched point by point. The inverse problem approach thereby avoids the ambiguities associated with the global quark-hadron duality ansatz and the effective threshold s_0 , resulting in smaller and controlled uncertainties.

Our results indicate that quark-hadron duality becomes applicable for $s \gtrsim 5 \text{ GeV}^2$, in agreement with the conclusions of several previous studies. Ref. [53] argues that a physically meaningful duality threshold s_0 should lie in a region where resonance structures are suppressed and perturbative QCD dominates. For example, the duality condition in $e^+e^- \rightarrow \text{hadrons}$ requires $s_0 \simeq (5.1\text{--}5.6) \text{ GeV}^2$, in remarkable agreement with the range of Λ identified in our analysis. Moreover, measurements of the R ratio by BESIII [54] are approximately flat and consistent with perturbative QCD from $s \simeq (2.2 \text{ GeV})^2$ up to the J/ψ resonance region, which coincides closely with the Λ -plateau region identified in our analysis. The quark-hadron duality also plays a central role in theoretical analyses of inclusive heavy-hadron decays and lifetimes. It is generally reliable for B -meson decays but can break down for D -meson decays, a pattern naturally understood from $m_b^2 > \Lambda$ whereas $m_c^2 < \Lambda$.

To provide a more direct and quantitative measure of quark-hadron duality, we evaluate the relative deviation of the continuum spectrum $\rho^c(s)$ from its perturbative counterpart $\rho^{\text{pert}}(s)$, for $s \geq 5 \text{ GeV}^2$. Specifically, we quantify this deviation through the ratio of norms: $\|\rho^c - \rho^{\text{pert}}\|_{L^2(5, \Lambda)} / \|\rho^{\text{pert}}\|_{L^2(5, \Lambda)}$. As shown in Fig. 6, this ratio lies between 2% and 4%, indicating that quark-hadron duality violations

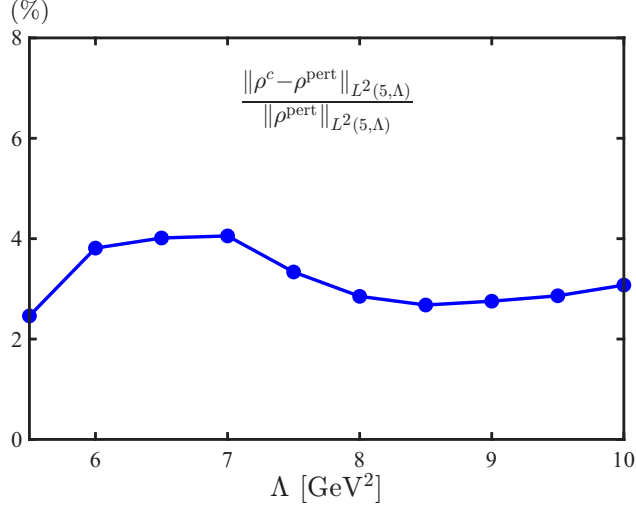


Figure 6: Quantitation of the quark-hadron duality.

are at $(2 - 4)\%$ level in this region. This result is in good agreement with Shifman's estimate of a 3% quark-hadron duality violation [14].

Such a level of duality violation is reasonable in the range $2 \text{ GeV} \lesssim \sqrt{s} \lesssim 3 \text{ GeV}$, where an effect below 5% is consistent with expectations. As emphasized in the preceding section, smooth agreement between the reconstructed continuum and the perturbative spectrum near the upper endpoint provides an important criterion for assessing the reliability of the reconstruction. The present analysis quantifies this criterion by requiring a finite interval over which the two spectra approach each other smoothly, with quark-hadron duality violations remaining below approximately 5%. We stress, however, that this level of duality violation is not imposed to determine any parameters, but is used solely as a diagnostic of the quality of the reconstructed solution.

4.2.2 Borel Transformation

To reduce the sensitivity to the quark-hadron duality ansatz and improve the accuracy of the duality approximation, QCD sum rules commonly employ a Borel transformation. Upon applying this transformation, the duality relation becomes

$$\int_{s_{\text{th}}}^{\infty} ds \rho^c(s) e^{-s/M^2} \simeq \frac{1}{\pi} \int_{s_0}^{\infty} ds \text{Im} \Pi^{\text{pert}}(s) e^{-s/M^2}, \quad (29)$$

where the Borel parameter M^2 must be determined within an appropriate window. The Borel window is determined by the simultaneous requirements of OPE convergence and ground-state pole dominance. In practice, these requirements are implemented by prescribing two numerical criteria for the relative sizes of the relevant contributions. On the one hand, the contribution from excited states and the continuum is required to remain sufficiently small relative to the ground-state pole, ensuring pole dominance and reducing sensitivity to the quark-hadron duality ansatz. On the other hand, the condensate contribution is required to remain sufficiently small to ensure the convergence and reliability of the OPE. However, the specific percentages assigned to these two criteria are not uniquely determined, and their choices inevitably introduce a degree of arbitrariness and potential bias. Moreover, whether the pole contribution itself constitutes a physically meaningful quantity remains a matter of debate. Indeed, its value is highly sensitive to the unphysical Borel parameter M^2 . Recent studies have sought to reduce this arbitrariness within the QCD sum rules framework, for example through improved moment sum rules [55] and Bayesian approaches based on the maximum entropy method [56].

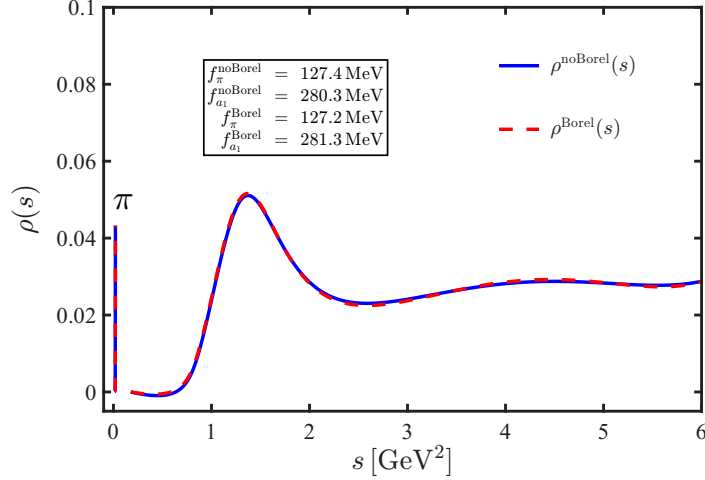


Figure 7: The Borel parameter is selected $M^2 \in [5, 80]$.

In this work, we also formulate and solve the inverse problem based on the Borel-transformed integral equation,

$$\int_{t_{\min}}^{\Lambda} ds e^{-s/M^2} \rho(s) = \int_0^{\Lambda} ds e^{-s/M^2} \rho^{\text{pert}}(s) + \Pi^{\text{cond}}(M^2), \quad (30)$$

with the corresponding results shown in Fig. 7. As can be seen, applying the Borel transformation within the inverse-problem framework has only a negligible effect on the reconstructed result. The analysis proceeds analogously to that based on the dispersion-relation integral equation, with the integral kernel replaced by e^{-s/M^2} . Rather than searching for a stability plateau as a function of M^2 , we employ a finite interval in M^2 as the input domain for the inverse-problem reconstruction. Specifically, we take $M^2 \in [5, 80]$ GeV², and find clear stability plateaus under independent variations of both the lower and upper limits. The upper limit may be chosen arbitrarily large and can, in principle, be extended to infinity, whereas the lower limit is constrained by the requirement that the OPE remain reliable. These two limits play roles analogous to q_{low}^2 and q_{up}^2 in the dispersion-relation formulation, respectively. However, we do not perform a detailed uncertainty analysis here, as the purpose of this comparison is to demonstrate the consistency between the results obtained with and without the Borel transformation.

In summary, the Borel transformation is not required in the inverse problem approach, since the excited states and continuum are reconstructed explicitly, while quark-hadron duality emerges dynamically in the perturbative region. Consequently, there is no need to enforce the suppression of these contributions through the Borel transformation. Moreover, we impose no prescribed numerical criteria on the relative sizes of the condensate and pole contributions, thereby avoiding the arbitrariness inherent in such criteria. The entire analysis is based on rigorously formulated mathematical framework for the inverse problem, without introducing any additional artificial conditions.

4.3 Vacuum Condensates

In this subsection, we can turn to discuss the respective contributions of the vacuum condensates to the ground state, excited states, and continuum spectrum.

Within QCD sum rules, the condensate contributions are effectively attributed entirely to the ground-state pole, because the excited states and continuum are dropped by the quark-hadron duality approximation. By contrast, the inverse problem approach resolves the ground state, excited states, and continuum spectrum simultaneously. The linearity of the inverse problem permits a decomposition of the reconstructed solution. By solving the corresponding linear system separately for the perturbative term and each condensate contribution in the input, we can quantify their individual contributions to the ground state, excited states, and continuum spectrum. The corresponding results are summarized in Table 3.

Table 3: Relative contributions of the perturbative and condensate terms in the OPE input to the reconstructed pion, $a_1(1260)$ -resonance, and continuum spectral weights.

	f_π^2 [GeV 2]	$f_{a_1(1260)}^2 \eta$ [GeV 2]	$\int_{s_{th}}^\Lambda ds \rho_i^c(s)$ [GeV 2]
Perturbative	0.01255 (77.3%)	0.07113 (113.8%)	0.08280 (94.4%)
$m_q \langle \bar{q}q \rangle$	-0.00028 (-1.7%)	0.00061 (1.0%)	-0.00034 (-0.4%)
$\langle \alpha_s GG \rangle$	0.00313 (19.2%)	-0.00694 (-11.1%)	0.00380 (4.3%)
$m_q \langle \bar{q}Gq \rangle$	0.00002 (0.1%)	-0.00005 (-0.1%)	0.00003 (0.0%)
$\langle \bar{q}q\bar{q}q \rangle$	0.00083 (5.1%)	-0.00227 (-3.6%)	0.00140 (1.6%)
Sum	0.016 (100.0%)	0.062 (100.0%)	0.088 (100.0%)

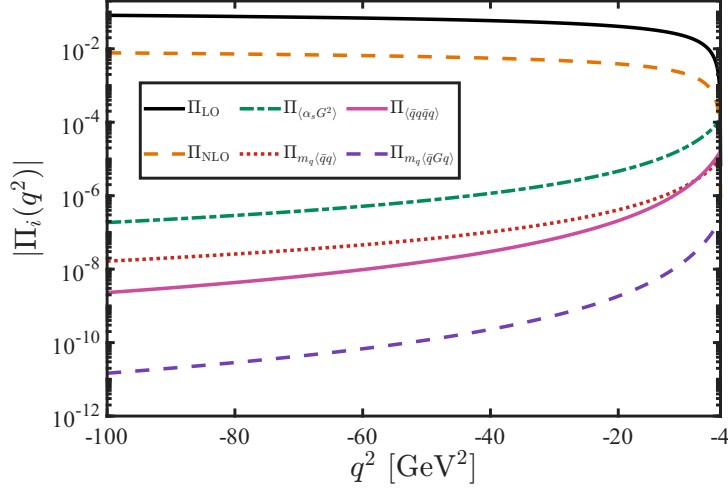


Figure 8

Table 3 presents the fractional contributions of the individual components of the theoretical input to the final reconstructed solution, corresponding to that shown in Fig. 4(e). The three numerical columns refer, respectively, to the pion-pole, $a_1(1260)$ -resonance, and continuum terms in

$$\int_{t_{\min}}^\Lambda ds \rho(s) = f_\pi^2 + f_{a_1(1260)}^2 \eta + \int_{s_{th}}^\Lambda ds \rho^c(s), \quad (31)$$

with $\eta \equiv \int_{s_{th}}^\Lambda ds \mathbb{B}\mathbb{W}(s, M_{a_1(1260)}, \Gamma_{a_1(1260)}) = 0.80$. Each row corresponds to an individual term in the OPE input of Eq. (7). In performing this decomposition, the regularization parameter α is fixed at the value determined in Fig. 4(e). The values in (parentheses) denote the fractional contribution of each input term to f_π^2 , $f_{a_1(1260)}^2 \eta$, and $\int_{s_{th}}^\Lambda ds \rho^c(s)$, respectively. We can see that, for each of the three components of the reconstructed solution, the perturbative input provides the dominant contribution, followed successively by the gluon and four-quark condensates. This hierarchy also provides direct evidence for the convergence of the OPE. The quark and mixed quark-gluon condensate contributions are small because the corresponding terms in the OPE input are themselves numerically suppressed, as seen in Fig. 8. Moreover, the Table 3 clearly shows that the condensate terms contribute not only to the ground-state pion residue f_π^2 , but also to the excited-state and continuum components. By contrast, in QCD sum rules, the excited states and continuum are dropped by the quark-hadron duality ansatz, so that the effects of all condensates in the input are effectively attributed to the ground-state. This provides another important distinction between the inverse problem approach and QCD sum rules.

As seen in Fig. 8, the NLO correction is larger than the gluon-condensate term in the input; nevertheless, Table 3 shows that the latter has a substantially greater impact on the reconstructed non-perturbative spectrum, including the pion pole, the excited state, and the continuum. This comparison demonstrates

that the impact of an input component on the inverse reconstruction is determined not by its magnitude alone, but also by its characteristic Q^2 dependence. The distinct Q^2 behavior of the condensate therefore allows them to play a disproportionately important role in shaping the non-perturbative spectrum.

Furthermore, Table 3 shows that, for each condensate, the contributions to f_π^2 , $f_{a_1(1260)}^2 \eta$, and $\int_{s_{\text{th}}}^\Lambda ds \rho^c(s)$ sum to zero within numerical accuracy. This cancellation is a noteworthy yet readily understood feature. To elucidate the cancellation among the pole and continuum contributions, we exploit the linearity of the dispersion relation and isolate the signed spectral component $\rho_i(s)$ associated with a given condensate term $\Pi_i(q^2)$. The once-subtracted dispersion relation is

$$\frac{\Pi_i(q^2) - \Pi_i(\sigma_0)}{q^2 - \sigma_0} = \int_0^\infty \frac{\rho_i(s)}{(s - q^2)(s - \sigma_0)} ds. \quad (32)$$

For large spacelike q^2 , the dispersive side becomes

$$\frac{\Pi_i(q^2) - \Pi_i(\sigma_0)}{q^2 - \sigma_0} = -\frac{\mathcal{A}_i}{q^2} - \frac{\mathcal{M}_i + \sigma_0 \mathcal{A}_i}{(q^2)^2} + o((q^2)^{-2}), \quad (33)$$

where

$$\mathcal{A}_i = \int_0^\infty \frac{\rho_i(s)}{s - \sigma_0} ds, \quad \mathcal{M}_i = \int_0^\infty \rho_i(s) ds. \quad (34)$$

The superconvergence condition does not require $\Pi_i(q^2)$ to be an exact inverse power of q^2 . It is sufficient that the OPE contain no term of order $1/q^2$, namely,

$$\lim_{q^2 \rightarrow -\infty} q^2 \Pi_i(q^2) = 0. \quad (35)$$

Under this condition, the OPE side of Eq. (32) has the asymptotic form

$$\frac{\Pi_i(q^2) - \Pi_i(\sigma_0)}{q^2 - \sigma_0} = -\frac{\Pi_i(\sigma_0)}{q^2} - \frac{\sigma_0 \Pi_i(\sigma_0)}{(q^2)^2} + o((q^2)^{-2}). \quad (36)$$

Matching Eqs. (33) and (36) gives

$$\mathcal{A}_i = \Pi_i(\sigma_0), \quad \mathcal{M}_i = 0. \quad (37)$$

The first equality is the subtraction constraint, while the second is the superconvergence relation resulting from the absence of a dimension-two contribution.

Decomposing the spectral component into the pion pole, the a_1 contribution, the low-energy continuum, and the high-energy OPE tail, one obtains

$$0 = f_{\pi,i}^2 + f_{a_1,i}^2 \int_0^\Lambda BW_1(s) ds + \int_0^\Lambda \rho_i^c(s) ds + \int_\Lambda^\infty \rho_i^{\text{OPE}}(s) ds. \quad (38)$$

For pure power corrections, the contact terms at $s = 0$ have a vanishing zeroth moment and no smooth high-energy spectral tail is present. Consequently, the first three terms in Eq. (38) cancel. For the logarithmic four-quark contribution, however, $\rho_{4q}^{\text{OPE}}(s) \propto s^{-3}$, and the high-energy tail must be retained:

$$f_{\pi,4q}^2 + f_{a_1,4q}^2 \int_0^\Lambda BW_1(s) ds + \int_0^\Lambda \rho_{4q}^c(s) ds = - \int_\Lambda^\infty \rho_{4q}^{\text{OPE}}(s) ds. \quad (39)$$

5 Summary and Outlook

The inverse problem approach constitutes a well-established branch of mathematics. In recent years, its application to determining non-perturbative QCD quantities has attracted sustained and growing interest in particle physics. In this work, we employ an inverse problem framework of dispersion relation to rigorously determine the decay constants and spectral properties of the pion and a_1 states, establishing a mathematically precise quantification without reliance on quark-hadron duality or Borel transformations. Our results not only align with those from recent lattice studies and other theoretical approaches, but more significantly, our method enables a systematic improvement in prediction accuracy. This is achieved through the incorporation of precise inputs, such as higher-order perturbative corrections and higher-dimensional condensates, together with the imposition of more physically motivated prior constraints. With these prospective refinements, our prediction for the spectral function $\rho(s)$ can be validated through comparisons with τ decay data in the region $s < M_\tau^2$ [46–49], as well as through the well-known Weinberg sum rules for $\rho_V(s) - \rho_A(s)$ [57],

$$\begin{aligned} \int_0^\infty ds [\rho_V(s) - \rho_A(s)] &= f_\pi^2, \\ \int_0^\infty ds s [\rho_V(s) - \rho_A(s)] &= 0, \end{aligned}$$

where $\rho_A(s)$ denotes the axial-vector spectral density and $\rho_V(s)$ corresponds to the vector spectral density [58, 59]. In addition, our theoretical framework not only yields physically reasonable decay constants but also naturally generates the resonance structure ($a_1(1260)$ and $a_1(1640)$). This facilitates the reconstruction of complete spectral functions, which are essential for determining the LECs of chiral perturbation theory.

It is worth emphasizing that the methodology employed in this work can be naturally extended to studies of heavy mesons, quarkonia, and baryon spectroscopy, enabling the extraction of corresponding physical quantities such as decay constants. These universal decay constants serve as the non-perturbative inputs in factorization theorems, thus playing a crucial role in achieving precise theoretical predictions. Collectively, all these advances will ultimately deepen our understanding of the diverse facets of strong interactions.

Acknowledgments

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A Proof of Uniqueness and Instability of Inverse Problem with a Pole

Then the dispersion-relation inverse problem with a pole is written as

$$\frac{f^{\text{pole}}}{m_\pi^2 - q^2} + \int_{s_{\text{th}}}^\Lambda ds \frac{f^c(s)}{s - q^2} = g(q^2), \quad (40)$$

where $q^2 \in [q_{\text{low}}^2, q_{\text{up}}^2]$ with $q_{\text{low}}^2 < q_{\text{up}}^2 \ll 0$ and $\Lambda > s_{\text{th}} > 0$.

Proof of uniqueness. It suffices to consider the homogeneous equation

$$\frac{f^{\text{pole}}}{m_\pi^2 - q^2} + \int_{s_{\text{th}}}^\Lambda ds \frac{f^c(s)}{s - q^2} = 0. \quad (41)$$

Let $u = q^2 - m_\pi^2$, $t = s - m_\pi^2$, $a = s_{\text{th}} - m_\pi^2$, $b = \Lambda - m_\pi^2$, and $h(t) = f_c(t + m_\pi^2)$. Since $0 < m_\pi^2 < s_{\text{th}}$, one has $0 < a < b$. The equation becomes

$$\frac{f^{\text{pole}}}{u} + \int_a^b dt \frac{h(t)}{u-t} = 0. \quad (42)$$

Writing $h(t) = F(t)/t$, we obtain

$$\frac{f^{\text{pole}}}{u} + \int_a^b dt \frac{F(t)}{t(u-t)} = 0. \quad (43)$$

For $|u| > b$,

$$\frac{1}{u-t} = \frac{1}{u} \sum_{k=0}^{\infty} \left(\frac{t}{u}\right)^k. \quad (44)$$

Therefore,

$$f^{\text{pole}} + \int_a^b dt \frac{F(t)}{t} + \sum_{k=0}^{\infty} \frac{1}{u^{k+1}} \int_a^b dt t^k F(t) = 0. \quad (45)$$

Since this identity holds on a non-degenerate interval of u , all coefficients vanish:

$$f^{\text{pole}} + \int_a^b dt \frac{F(t)}{t} = 0, \quad \int_a^b dt t^k F(t) = 0, \quad k = 0, 1, 2, \dots \quad (46)$$

The second relation implies that F is orthogonal to all polynomials. By the density of polynomials in $L^2(a, b)$, we have

$$F(t) = 0 \quad \text{a.e. on } [a, b]. \quad (47)$$

Since $t \geq a > 0$, this gives $h(t) = F(t)/t = 0$, and hence $f^c(s) = 0$ a.e. on $[s_{\text{th}}, \Lambda]$. The first relation then implies $f^{\text{pole}} = 0$. Therefore, the homogeneous equation has only the trivial solution, and the solution is unique. $\square$

Proof of instability. We construct a counterexample. Let $L = \Lambda - s_{\text{th}}$. For an exact solution (f^{pole}, f_c) , define

$$f^{\text{pole},n} = f^{\text{pole}}, \quad f^{c,n}(s) = f^c(s) + \sqrt{n} \cos\left(\frac{n\pi(s - s_{\text{th}})}{L}\right). \quad (48)$$

Then

$$\|(f^{\text{pole},n}, f^{c,n}) - (f^{\text{pole}}, f^c)\|_{L^2}^2 = \left[\int_{s_{\text{th}}}^{\Lambda} ds n \cos^2\left(\frac{n\pi(s - s_{\text{th}})}{L}\right) \right]^{1/2} = \sqrt{\frac{nL}{2}} \rightarrow \infty. \quad (49)$$

Let g_n be the data corresponding to $(f^{\text{pole},n}, f^{c,n})$. Since the pole part is unchanged,

$$g_n(q^2) - g(q^2) = \int_{s_{\text{th}}}^{\Lambda} ds \frac{\sqrt{n} \cos\left(\frac{n\pi(s - s_{\text{th}})}{L}\right)}{s - q^2}. \quad (50)$$

With $\kappa_n = n\pi/L$, integration by parts gives

$$g_n(q^2) - g(q^2) = \frac{\sqrt{n}}{\kappa_n} \int_{s_{\text{th}}}^{\Lambda} ds \frac{\sin(\kappa_n(s - s_{\text{th}}))}{(s - q^2)^2}, \quad (51)$$

where the boundary term vanishes because $\sin 0 = \sin(n\pi) = 0$. Since $q^2 \leq q_{\text{up}}^2 < 0 < s_{\text{th}}$, we have

$$s - q^2 \geq s_{\text{th}} - q_{\text{up}}^2 =: d > 0. \quad (52)$$

Therefore,

$$\|g_n(q^2) - g(q^2)\|_{L^2}^2 \leq \frac{\sqrt{n}}{\kappa_n} \int_{s_{\text{th}}}^{\Lambda} \frac{ds}{d^2} = \frac{L^2}{\pi d^2} \frac{1}{\sqrt{n}}. \quad (53)$$

It follows that

$$\|g_n - g\|_{L^2}^2 \leq \frac{L^2}{\pi d^2} \frac{\sqrt{q_{\text{up}}^2 - q_{\text{low}}^2}}{\sqrt{n}} \rightarrow 0. \quad (54)$$

Thus $g_n \rightarrow g$, while the corresponding solutions diverge in norm. Hence the inverse problem is unstable. $\square$

References

- [1] D. J. Gross and F. Wilczek. “Asymptotically Free Gauge Theories - I”. *Phys. Rev. D* 8 (1973) .
- [2] D. J. Gross and F. Wilczek. “Ultraviolet Behavior of Nonabelian Gauge Theories”. *Phys. Rev. Lett.* 30 (1973) .
- [3] H.-n. Li and H. Umeeda. “Vacuum polarization contribution to muon $g - 2$ as an inverse problem”. *Phys. Rev. D* 102.9 (2020) . arXiv: 2004.06451 [hep-ph].
- [4] Z. Xing, L. Chang, and K. Raya. “Electromagnetic structure of axial-vector mesons and implications for the muon $g - 2$ ” (Sept. 2025) (Sept. 2025) . arXiv: 2509.17311 [hep-ph].
- [5] A. Bazavov et al. “Hadronic vacuum polarization for the muon $g-2$ from lattice QCD: Complete short and intermediate windows”. *Phys. Rev. D* 111.9 (2025) . arXiv: 2411.09656 [hep-lat].
- [6] R. Aliberti et al. “The anomalous magnetic moment of the muon in the Standard Model: an update” (May 2025) (May 2025) . arXiv: 2505.21476 [hep-ph].
- [7] M. Ubiali. “Parton Distribution Functions and Their Impact on Precision of the Current Theory Calculations”. In: Apr. 2024. arXiv: 2404.08508 [hep-ph].
- [8] M. Mai et al. “Three-Body Dynamics of the $a_1(1260)$ Resonance from Lattice QCD”. *Phys. Rev. Lett.* 127.22 (2021) . arXiv: 2107.03973 [hep-lat].
- [9] H. Yan et al. “ ω Meson from Lattice QCD”. *Phys. Rev. Lett.* 133.21 (2024) . arXiv: 2407.16659 [hep-lat].
- [10] M. T. Hansen and S. R. Sharpe. “Lattice QCD and Three-particle Decays of Resonances”. *Ann. Rev. Nucl. Part. Sci.* 69 (2019) . arXiv: 1901.00483 [hep-lat].
- [11] M. Ding, C. D. Roberts, and S. M. Schmidt. “Emergence of Hadron Mass and Structure”. *Particles* 6.1 (2023) . arXiv: 2211.07763 [hep-ph].
- [12] M. A. Shifman, A. I. Vainshtein, and V. I. Zakharov. “QCD and Resonance Physics. Theoretical Foundations”. *Nucl. Phys. B* 147 (1979) .
- [13] M. A. Shifman, A. I. Vainshtein, and V. I. Zakharov. “QCD and Resonance Physics: Applications”. *Nucl. Phys. B* 147 (1979) .
- [14] M. A. Shifman. “Quark hadron duality”. In: *8th International Symposium on Heavy Flavor Physics*. Vol. 3. Singapore: World Scientific, July 2000, pp. 1447–1494. arXiv: hep-ph/0009131.
- [15] H.-n. Li and H. Umeeda. “QCD sum rules with spectral densities solved in inverse problems”. *Phys. Rev. D* 102 (2020) . arXiv: 2006.16593 [hep-ph].
- [16] D. B. Leinweber. “QCD sum rules for skeptics”. *Annals Phys.* 254 (1997) . arXiv: nucl-th/9510051.
- [17] S. Narison. “QCD spectral sum rules 2022”. *Nucl. Part. Phys. Proc.* 324-329 (2023) . arXiv: 2211.14536 [hep-ph].
- [18] P. Colangelo and A. Khodjamirian. “QCD sum rules, a modern perspective”. (Oct. 2000) . arXiv: hep-ph/0010175.

- [19] Z.-X. Zhao, Y.-P. Xing, and R.-H. Li. “A progress in the inverse matrix method in QCD sum rules”. *Eur. Phys. J. C* 84.10 (2024) . arXiv: 2407.09819 [hep-ph].
- [20] H.-N. Li et al. “ D meson mixing as an inverse problem”. *Phys. Lett. B* 810 (2020) . arXiv: 2001.04079 [hep-ph].
- [21] A.-S. Xiong et al. “Inverse Problem Approach for Non-Perturbative QCD: Theoretical Foundation” (Nov. 2022) (Nov. 2022) . arXiv: 2211.13753 [hep-th].
- [22] J. F. Donoghue and E. Golowich. “Chiral sum rules and their phenomenology”. *Phys. Rev. D* 49 (1994) . arXiv: hep-ph/9307262.
- [23] D. Boito et al. “Low-energy constants from ALEPH hadronic tau decay data”. *PoS CD15* (2016) . arXiv: 1509.08125 [hep-ph].
- [24] D. Boito et al. “Low-energy constants and condensates from ALEPH hadronic τ decay data”. *Phys. Rev. D* 92.11 (2015) . arXiv: 1503.03450 [hep-ph].
- [25] A. N. Tikhonov and V. Y. Arsenin. *Solutions of ill-posed problems*. John Wiley and Sons, New York, 1977.
- [26] V. L. Chernyak and A. R. Zhitnitsky. “Asymptotic Behavior of Exclusive Processes in QCD”. *Phys. Rept.* 112 (1984) .
- [27] K.-C. Yang. “Light-cone distribution amplitudes of axial-vector mesons”. *Nucl. Phys. B* 776 (2007) . arXiv: 0705.0692 [hep-ph].
- [28] J.-J. Han et al. “Establishing CP Violation in b-Baryon Decays”. *Phys. Rev. Lett.* 134.22 (2025) . arXiv: 2409.02821 [hep-ph].
- [29] J.-J. Han et al. “CP violation in two-body hadronic Λ_b decays in the PQCD approach”. *Phys. Rev. D* 112.5 (2025) . arXiv: 2506.07197 [hep-ph].
- [30] Y. Zheng et al. “Invisible and semi-invisible decays of bottom baryons*”. *Chin. Phys. C* 48.8 (2024) . arXiv: 2404.04337 [hep-ph].
- [31] J. Lu et al. “Exclusive production of double light neutral mesons at the e+e- colliders”. *Phys. Rev. D* 109.9 (2024) . arXiv: 2403.04762 [hep-ph].
- [32] L.-Y. Li et al. “ $\Lambda_b \rightarrow P \ell$ factorization in QCD”. *Phys. Rev. D* 109.11 (2024) . arXiv: 2401.11978 [hep-ph].
- [33] Y. Ji et al. “Next-to-Next-to-Leading-Order QCD Prediction for the Pion Form Factor”. *Phys. Rev. Lett.* 134.22 (2025) . arXiv: 2411.03658 [hep-ph].
- [34] J. Gao et al. “Next-to-Next-to-Leading-Order QCD Prediction for the Photon-Pion Form Factor”. *Phys. Rev. Lett.* 128.6 (2022) . arXiv: 2106.01390 [hep-ph].
- [35] J. Bijnens and G. Ecker. “Mesonic low-energy constants”. *Ann. Rev. Nucl. Part. Sci.* 64 (2014) . arXiv: 1405.6488 [hep-ph].
- [36] J. Bijnens, N. Hermansson-Truedsson, and A. Rodríguez-Sánchez. “The Pion Sum Rule beyond the Chiral Limit” (Feb. 2026) (Feb. 2026) . arXiv: 2602.05672 [hep-ph].
- [37] T. Zhong et al. “Improved light-cone harmonic oscillator model for the pionic leading-twist distribution amplitude”. *Phys. Rev. D* 104.1 (2021) . arXiv: 2102.03989 [hep-ph].
- [38] H.-n. Li. “Dispersive derivation of the pion distribution amplitude”. *Phys. Rev. D* 106.3 (2022) . arXiv: 2205.06746 [hep-ph].
- [39] S. Narison. “Mini-review on QCD spectral sum rules”. *Nucl. Part. Phys. Proc.* 258-259 (2015) . arXiv: 1409.8148 [hep-ph].
- [40] A. Khodjamirian. *Hadron Form Factors*. CRC Press, Apr. 2020.

- [41] D. M. Asner et al. “Hadronic structure in the decay $\tau \rightarrow \tau\text{-neutrino } \pi^- \pi^0 \pi^0$ and the sign of the tau-neutrino helicity”. *Phys. Rev. D* 61 (2000) . arXiv: [hep-ex/9902022](#).
- [42] J.-F. Jiang and S.-L. Zhu. “Radial excitations of mesons and nucleons from QCD sum rules”. *Phys. Rev. D* 92.7 (2015) . arXiv: [1508.00677 \[hep-ph\]](#).
- [43] S. Navas et al. “Review of particle physics”. *Phys. Rev. D* 110.3 (2024) .
- [44] M. G. Bowler. “The A1 Revisited”. *Phys. Lett. B* 182 (1986) .
- [45] R. Aaij et al. “Studies of the resonance structure in $D^0 \rightarrow K^\mp \pi^\pm \pi^\pm \pi^\mp$ decays”. *Eur. Phys. J. C* 78.6 (2018) . arXiv: [1712.08609 \[hep-ex\]](#).
- [46] S. Schael et al. “Branching ratios and spectral functions of tau decays: Final ALEPH measurements and physics implications”. *Phys. Rept.* 421 (2005) . arXiv: [hep-ex/0506072](#).
- [47] A. Pich. “Precision Tau Physics”. *Prog. Part. Nucl. Phys.* 75 (2014) . arXiv: [1310.7922 \[hep-ph\]](#).
- [48] M. Davier et al. “Update of the ALEPH non-strange spectral functions from hadronic τ decays”. *Eur. Phys. J. C* 74.3 (2014) . arXiv: [1312.1501 \[hep-ex\]](#).
- [49] Numerical spectral function files: <http://aleph.web.lal.in2p3.fr/tau/specfun13.html>.
- [50] D. Gomez Dumm, A. Pich, and J. Portoles. “ $\tau \rightarrow \pi^- \pi^+ \pi^- \nu(\tau)$ decays in the resonance effective theory”. *Phys. Rev. D* 69 (2004) . arXiv: [hep-ph/0312183](#).
- [51] Y. Aoki et al. “FLAG Review 2024” (Nov. 2024) (Nov. 2024) . arXiv: [2411.04268 \[hep-lat\]](#).
- [52] M. Wingate et al. “Properties of the a1 meson from lattice QCD”. *Phys. Rev. Lett.* 74 (1995) . arXiv: [hep-ph/9502274](#).
- [53] S. Narison. “QCD parameters and SM-high precision from e^+e^- to Hadrons: Updated”. *Nucl. Phys. A* 1046 (2024) . arXiv: [2402.13983 \[hep-ph\]](#).
- [54] M. Ablikim et al. “Measurement of the Cross Section for $e^+e^- \rightarrow \text{Hadrons}$ at Energies from 2.2324 to 3.6710 GeV”. *Phys. Rev. Lett.* 128.6 (2022) . arXiv: [2112.11728 \[hep-ex\]](#).
- [55] J.-P. Zhang, X.-L. Chen, and W. Chen. “An improved moment QCD sum rule” (July 2026) (July 2026) . arXiv: [2607.11536 \[hep-ph\]](#).
- [56] P. Gubler and M. Oka. “A Bayesian approach to QCD sum rules”. *Prog. Theor. Phys.* 124 (2010) . arXiv: [1005.2459 \[hep-ph\]](#).
- [57] S. Weinberg. “Precise relations between the spectra of vector and axial vector mesons”. *Phys. Rev. Lett.* 18 (1967) .
- [58] H.-n. Li. “Dispersive analysis of glueball masses”. *Phys. Rev. D* 104.11 (2021) . arXiv: [2109.04956 \[hep-ph\]](#).
- [59] M. Gonzalez-Alonso, A. Pich, and J. Prades. “Violation of Quark-Hadron Duality and Spectral Chiral Moments in QCD”. *Phys. Rev. D* 81 (2010) . arXiv: [1001.2269 \[hep-ph\]](#).